The Supply-Side Effects of Healthcare Privatization: Evidence from Puerto Rico*

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Abstract

We provide, what is to our knowledge, the first causal evidence on privatizing an entire public healthcare delivery system. In doing so, we study Puerto Rico's 1993 health reform which sold or leased government healthcare establishments to private providers and shifted payment and coverage to private managed-care organizations paid by capitation, changing ownership, payment, and coverage rules at once. Because the reform rolled out across all 78 counties between 1994 and 2000, we estimate its effects with the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021). Findings indicate that the healthcare sector contracted, where per thousand residents, employment fell 13.0 percent and Medicare-certified hospital beds 7.8 percent relative to their pre-reform means. The healthcare labor market adjustment was seen in employment levels rather than wages. Patient health did not deteriorate on average, and infant mortality fell by 13 to 27 percent of its pre-reform mean in the more urban, populous, and healthcare-intensive counties. A back-of-the-envelope calculation estimates the freed real resources at about $844 million in 2025 dollars over the four post-reform years under our most conservative estimate, enough to make net welfare positive before health gains.

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1 Introduction

Over recent decades, governments have increasingly opted to deliver care through the private sector. Whether this lowers costs without sacrificing quality is a question economists have long debated (Shleifer, 1998). Some evidence suggests that privatization increases firm productivity (Megginson & Netter, 2001), yet little is known about its effects on consumers. The concern is most severe in healthcare, where contracts are incomplete (Grossman & Hart, 1986), quality is hard to verify (Arrow, 1963; Dranove, Kessler, McClellan, & Satterthwaite, 2003), and competition is limited (Cutler & Scott Morton, 2013; Gaynor, Ho, & Town, 2015). Under these conditions, Hart, Shleifer, and Vishny (1997) argue that private ownership strengthens the incentive to cut costs, which can decrease the quality of care no contract can specify. The net effect on spending and health has been shown to be determined by three margins: who owns the facilities (Duggan, Gupta, Jackson, & Templeton, 2023), how providers are paid (Gruber, 2017), and who is granted access (Finkelstein, 2007). Because they pull in opposite directions, the net effect of moving them together (which is what privatizing a delivery system entails), cannot be inferred from research that studies each one at a time. We fill this gap by asking how privatizing an entire delivery system affects the per-capita supply of healthcare and population health by studying Puerto Rico's 1993 health reform, which changed all three margins at once.

Healthcare is among the largest claims on government budgets. In the United States, Medicare and Medicaid together account for more than a fifth of federal spending (Medicaid and CHIP Payment and Access Commission, 2026), and governments have turned to three variables to reduce that cost. The first is ownership: the government's share of U.S. hospital capacity fell by about 42 percent between 1983 and 2019 (Duggan et al., 2023), and systems such as the Department of Veterans Affairs and England's NHS increasingly buy care from private providers (Congressional Budget Office, 2021; Congressional Research Service, 2025a; Farmer, 2023; King's Fund, 2026; National Audit Office, 2026). The second is payment, a shift from fee-for-service to capitation. As of 2024, capitated Medicare Advantage plans enrolled 54 percent of eligible Medicare beneficiaries and accounted for about $462 billion of federal Medicare spending (Kaiser Family Foundation, 2024a), while capitated managed-care plans covered more than half of Medicaid benefit spending (Gruber, 2017; Kaiser Family Foundation, 2025; Medicaid and CHIP Payment and Access Commission, 2026; Shepard & Wallace, 2026). Neither has lowered costs. Medicare pays Advantage plans about 22 percent more than comparable enrollees would cost under fee-for-service (Medicare Payment Advisory Commission, 2024), and the shift of Medicaid enrollees into managed care did not, on average, reduce spending (Duggan & Hayford, 2013). The third is access: eligibility rules set both how many people a program covers and how much care it covers, and as of 2025 Medicaid and CHIP reached 78 million people, with Medicaid spending alone near $957 billion in fiscal year 2024, roughly two-thirds of it federal (Kaiser Family Foundation, 2026; Medicaid and CHIP Payment and Access Commission, 2026).

The evidence on each margin is itself mixed. For example, when it comes to ownership, shifting delivery from the public to the private sector has been found possibly worsen care. Chan, Card, and Taylor (2023) find that patients treated in Veterans Affairs hospitals survive at higher rates than those treated privately, and Goodair and Reeves (2022) link greater NHS outsourcing to higher treatable mortality. Yet in both, the public healthcare programs are kept as alternatives to patients and are never themselves privatized. The closest evidence to a true transfer, and the nearest to our own setting, follows a single facility sold to private owners. Such is the setting that Duggan et al. (2023) study, where they find that privatizing a public hospital cuts labor and raises mortality among elderly patients while private owners select the more profitable patients. But each facility is privatized within an intact market that absorbs its rejected patients and laid-off workers, so the estimates are local to a single facility and cannot speak to privatizing an entire system at once. On payment, moving from fee-for-service to capitated managed-care plans (while providers stay private) has lowered utilization in some settings, with contested effects on spending and quality (Agafiev Macambira, Geruso, Lollo, Ndumele, & Wallace, 2026; Aizer, Currie, & Moretti, 2007; Chorniy, Currie, & Sonchak, 2018; Curto, Einav, Finkelstein, Levin, & Bhattacharya, 2019; Dranove, Ody, & Starc, 2021; Duggan, 2004; Duggan, Gruber, & Vabson, 2018; Geruso, Layton, & Wallace, 2023; Marton, Yelowitz, & Talbert, 2014). On access, extending insurance coverage raises the demand for care (Finkelstein, 2007), a margin studied apart from ownership and payment. Since these studies predict contradicting outcomes, the joint effect of changing all three margins simultaneously remains to be empirically tested.

To this aim, we study Puerto Rico's 1993 health reform, a bundled policy that moved all three margins. As part of this policy, the Commonwealth sold or leased its public clinics and hospitals to private providers, replaced the payment of healthcare providers from salaried to capitated, and enrolled all income-eligible patients in health plans provided by Medicaid managed-care organizations. We study the reform across all 78 counties of the island, which transitioned in five annual cohorts between 1994 and 2000, with a staggered difference-in-differences design built on the not-yet-treated estimator of Callaway and Sant'Anna (2021) and administrative data on employment, hospital capacity, and vital statistics. This setting offers two advantages: the near-absence of cross-market spillovers and the exogeneity of funding. The first arises because Puerto Rico is an island, so the not-yet-treated counties carry none of the cross-border patient and provider flows that complicate mainland designs. Second, because federal Medicaid financing to Puerto Rico is capped rather than matched (Congressional Research Service, 2025b), unlike the open-ended federal match that U.S. states receive, the resource changes we measure do not reflect an incentive to draw down federal funds.

This paper makes three contributions. The first is to the literature on privatization of healthcare delivery and facilities ownership, where we provide (to our knowledge) the first causal evidence on privatizing an entire delivery system. The second contribution is to the literature on Medicaid managed care and the payer side of privatization, which studies how moving enrollees into capitated plans changes spending, plan choice, and the outcomes patients experience (Curto et al., 2019; Dranove et al., 2021; Duggan et al., 2018). That work observes the payer and the patient but not the provider, and so is silent on what privatization does to the supply of care itself. We document that effect by showing the per-capita reallocation of healthcare labor and capacity. Because we observe the quantity of care, our results show how privatized providers reallocate resources. The third contribution is to the study of how market structure shapes privatization's effects. The supply response was not uniform across the island, since it clustered in the most highly concentrated healthcare markets (where the public system had been largest and population is dense), a cross-market heterogeneity that neither single-facility settings, nor payer-side settings (which average over geography) can identify.

The paper proceeds as follows. Section 2 describes the reform and the pre-reform system, Section 3 the data, Section 4 the theoretical framework, Section 5 the empirical strategy, Section 6 the results, Section 7 the robustness analysis, and Section 8 concludes.

2 Setting

2.1 The Pre-Reform Public Delivery System

As of the mid 1950s, the Commonwealth of Puerto Rico delivered healthcare through an integrated public system built on the regionalized principles of Arbona and Ramirez de Arellano (1978) which had universal access at the point of service, a four-tier referral hierarchy, and direct public employment of clinical staff. The base tier comprised 78 Centers of Diagnostic and Treatment (CDTs), one per county, providing primary, prenatal, dental, and pharmaceutical care through salaried government clinicians; above them sat five Area Hospitals, six Regional Hospitals, and the supra-tertiary Puerto Rico Medical Center in the capital of San Juan. The system was the dominant source of care, where in 1990 it operated about 40 percent of the Commonwealth's hospital beds and served roughly 1.8 million residents, which corresponded to about 52 percent of the population (Instituto de Administración y Política de Salud, 1995).

The reform confronted a public delivery system bound by a hard budget constraint. By 1990 total health spending had reached $2.5 billion, about 11 percent of gross domestic product (World Health Organization & Pan American Health Organization, 1998), yet federal Medicaid funds arrived as a capped allotment, so the Commonwealth financed most public provision from its own budget (Kaiser Family Foundation, 2024b; Medicaid and CHIP Payment and Access Commission, 2020). According to Commonwealth of Puerto Rico (1993), this constraint bound hardest on the public sector because a two-tier structure had sorted residents by ability to pay: privately insured residents used private providers, while the public Department of Health remained the residual provider for a poorer, sicker, and adversely selected population. A fixed budget could not match that population's need for capital and medical technology, and the under-resourcing was visible in outcomes, with contemporaneous assessments documenting quantity and quality shortfalls relative to private providers and higher recorded mortality among the publicly served (World Health Organization & Pan American Health Organization, 1998). The enabling legislation attributed the decline in public-sector quality to inadequate budgets, the rising cost of medical technology, bureaucratic centralization, and political interference (Commonwealth of Puerto Rico, 1993). The reform's stated aim was to contract private insurers and providers under capitation to extend quality care to the low-income population, while reducing costs (Commonwealth of Puerto Rico, 1993).

2.2 The 1993 Reform

The Puerto Rico Health Insurance Administration Act (Act 72 of September 7, 1993) created a single public purchaser of coverage, the Administración de Seguros de Salud (ASES), and restructured delivery along all three margins at once (Commonwealth of Puerto Rico, 1993; World Health Organization & Pan American Health Organization, 1998).[1] Ownership

[1] Eligibility extended to residents with household incomes below 200 percent of the Puerto Rico poverty threshold, alongside the existing Medicaid-eligible population; the benefits package covered inpatient and outpatient hospital care, physician and ambulatory services, pharmacy, dental, mental-health and substance-

moved from public to private, where the 78 CDTs (one per county) and the 11 Area and Regional Hospitals were sold or placed under long-term lease to private providers, while the Commonwealth retained the supra-tertiary Medical Center (Instituto de Administración y Política de Salud, 1995). Healthcare providers payment moved from Department of Health salaries to capitation paid by ASES-contracted insurers at rates negotiated bilaterally with providers (Commonwealth of Puerto Rico, 1993; World Health Organization & Pan American Health Organization, 1998). Access moved from open entitlement at public facilities to enrollment in a single managed-care network (designated by county of residence) with primary-care gatekeeping and income-scaled cost-sharing. The public system had been an open outside option for the privately insured and the main one for the uninsured. Therefore, its transfer to private providers removed that fallback [2]. Financing details and the post-2002 policy changes such as the successor programs are reported in Appendix 10.

ASES divided the Commonwealth into eight administrative health regions and awarded each to a single managed-care organization through competitive bidding, creating regional monopolies, and rolled the regions out between February 1994 and July 2000 (Brugueras Fabre, 2002; World Health Organization & Pan American Health Organization, 1998) (see Figure 1 panel (a) for a map of the cohorts across all counties). The number of distinct managed-care organizations participating island-wide grew from two in 1995 to seven by 2005, with two present throughout the period and several smaller plans entering and exiting, and mental-health services were carved out to separate behavioral-health organizations beginning in 2002.[3] The order ran from the sparsely populated and outer-island regions to the capital

abuse treatment, and preventive services, without exclusions for pre-existing conditions. Cost-sharing was \$1 to \$4 per visit and \$0.50 to \$3 per prescription, scaled by income (Centers for Medicare and Medicaid Services, 2000, 2001, 2003; Commonwealth of Puerto Rico, 1993; Instituto de Administración y Política de Salud, 1995; World Health Organization & Pan American Health Organization, 1998).

[2] Since we lack data on the size of the new provider networks, we cannot say whether enrollees gained access to more providers or fewer.

[3] Counts of participating organizations by year (two in 1995; four in 1996; three in 1997; four in 1998 and 1999; three in 2000; four in 2001; six in 2002; five in 2003; and seven in 2004 and 2005) are compiled from the CMS Medicaid managed-care enrollment reports for 1995–1999 (Centers for Medicare and Medicaid Services, 1995, 1996, 1997, 1998, 1999) and the CMS national summaries of state Medicaid managed-care programs for 2000–2005 (Centers for Medicare and Medicaid Services, 2000, 2001, 2002, 2003, 2004, 2005); the 1995 report records two participating organizations without naming them.

of San Juan, so treatment timing is correlated with county urbanization, the identification concern we address in Section 5. We group the eight transition dates into five annual cohorts by year of transition: 1994 (10 counties), 1995 (12), 1996 (30), 1998 (25), and 2000 (San Juan alone). Because San Juan holds the largest concentration of healthcare establishments and employment, we report its influence separately and show that excluding it attenuates the supply-side magnitude (Section 7).

As the regions implemented the policy, Medicaid managed care beneficiaries rose from roughly 50,000 residents in 1994 to about 1.64 million by fiscal year 2000, 42.8 percent of the population, while its outlay scaled from essentially zero to about $1.9 billion in 2025 dollars (about 5 percent of the Commonwealth's consolidated budget),[4] most of it premium pass-through to the contracting managed-care organizations rather than direct public provision (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1999).[5] During this time, the government decreased its total spending in healthcare, while patients faced rising costs for medical services, a trend that showed a faster growth rate than healthcare workers average wages (see Figure 1, Panel (c)[6] and Panel (d)[7]).

[4]All shares here are computed in nominal fiscal-year-2000 dollars. Puerto Rico's consolidated budget that year was $19.9 billion, of which the General Fund was $7.0 billion (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1999). ASES's $1.0 billion was about 5 percent of the consolidated budget; public health overall (ASES plus the $0.9 billion Department of Health, some $1.9 billion) was about 10 percent.

[5]In nominal terms the outlay was about $1.0 billion in fiscal year 2000; the text deflates it to 2025 dollars with the U.S. CPI-U. Enrollment is read from the published budget figure and is approximate. The implied program cost averaged about $610 per enrollee per year ($51 per member per month) in nominal fiscal-year-2000 dollars, a descriptive average of appropriations over enrollment rather than a negotiated capitation rate.

[6]Panel (c) is in constant 2025 dollars, deflated with the U.S. CPI-U. Public delivery is the cost of the government-operated agencies (the Assistance Fund for Health Services, AFASS; the Medical Center, ASEM; and the Cardiovascular Center), reported on a financial-statements basis that runs below the operating-budget totals used elsewhere; private delivery is the capitated premiums the government pays the managed-care organizations through ASES, not out-of-pocket or private-insurance spending; and total government health spending is their sum. The household medical-care share on the right axis is a national-accounts share of total personal consumption, not out-of-pocket spending, shown for comparison only and not comparable to the dollar levels. Sources: the Commonwealth budget volumes (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1997, 1999, 2001; Estado Libre Asociado de Puerto Rico, Oficina de Presupuesto y Gerencia, 1989, 1993; Gobierno de Puerto Rico, 2006) for spending and Junta de Planificación de Puerto Rico (2024) for the consumption share.

[7]The real wage per healthcare worker in Panel (d) is the annual QCEW wage bill, total quarterly wages summed over the four calendar quarters, divided by annual average employment for NAICS 621–623 in Puerto Rico, in constant 2025 dollars (U.S. CPI-U); it is a covered-wage measure that excludes self-employed

3 Data

The analysis combines six data sources, all used at the county level and at annual frequency within 1990–2001[8]. We measure the local healthcare sector in two ways. Employment comes from the U.S. Bureau of Labor Statistics Quarterly Census of Employment and Wages (QCEW) (U.S. Bureau of Labor Statistics, 2025), which reports county-quarter employment, wages, and establishment counts by NAICS code; the healthcare sector is the union of ambulatory care (NAICS 621), hospitals (622), and nursing and residential care (623), and we report NAICS 621 separately because it maps most directly onto the privatized county clinic network. Because the QCEW counts establishments of every ownership type, transferring facilities from public to private does not by itself move workers into or out of the measure, since a change in employment is a change in the number of healthcare jobs, not a reclassification across ownership categories. Hospital capacity comes from the Centers for Medicare and Medicaid Services Provider of Services (POS) files (Centers for Medicare and Medicaid Services, 2025), aggregated to the county-year. The two measures see different parts of the system: the POS files identify only Medicare-certified providers and so exclude the 78 CDTs, whose primary care did not require certification. The privatized clinic layer that was the reform's main target therefore shows up in QCEW employment, not in the bed count, which corroborates the hospital tier.

The Puerto Rico Department of Health's Vital Statistics reports (Puerto Rico Department of Health, 2025) supply the patient-health outcomes (the crude birth rate, infant mortality, the low-birth-weight share, and the government and private hospital-birth shares), along with county population and the crude net migration rate; the Department issued these only as unstructured annual PDFs, which we digitized, reconciled across years, and assembled into the structured county-year panel used here. The worker-margin outcomes

physician income and managed-care-organization margins. Fiscal years run July–June and are labeled by their ending year, whereas the QCEW wage is a calendar-year quantity, so the two are aligned by year label. Sources: Junta de Planificación de Puerto Rico (2024) (Table 5) and the U.S. Bureau of Labor Statistics Quarterly Census of Employment and Wages (U.S. Bureau of Labor Statistics, 2025).

[8]Except for the (Centers for Medicare and Medicaid Services, 2025), which is only used as of 1991)

(labor-force participation, the employment-to-population ratio, and unemployment) come from the monthly county labor-force survey of the Puerto Rico Department of Labor and Human Resources (Puerto Rico Department of Labor and Human Resources, 2025). Two further sources support the robustness checks: the National Center for Health Statistics National Vital Statistics System (NVSS) natality files (National Center for Health Statistics, 2025) corroborate the county birth outcomes with individual records, and the Behavioral Risk Factor Surveillance System (BRFSS) (Centers for Disease Control and Prevention, 2025) supplies the individual adult coverage reports we use to test the access first stage. Appendix 10 gives the construction details.

The Vital Statistics population series (the Commonwealth's official population count) is the denominator for all per-thousand-resident outcomes. Per-capita scaling places the labor, supply, and health outcomes aligned with the literature (Hackmann, Heining, Klimke, Polyakova, & Seibert, 2025; Wallace, 2023). The balanced QCEW healthcare panel covers 59 counties (76% of the total) and the balanced CMS panel 40 (51% of the total), while the Vital Statistics and labor-force panels covers 78 (100% of the total); the QCEW censorship reflects the Bureau of Labor Statistics' suppression of county-industry data with few establishments or employment, which tilts the balanced healthcare sample towards counties with a higher health concentrated market. The supply-side estimates are therefore identified on a sample that already over-weights population dense counties, so their concentration in the metropolitan areas should be read as consistent with both heterogeneity and sample composition, a caveat we address in Section 7.

Table 1 reports summary statistics, and Table 2 the cohort-specific pre-reform (1990–1993) means of population, public-system intensity, healthcare and placebo-industry employment, labor-market activity, demographic flows, and patient-health covariates, with one-way ANOVA p-values for the joint null of equal cohort means. The cohorts differ systematically in urban intensity, reflecting the rural-to-urban rollout.

4 Theoretical Framework

Ownership and payment each work to lower per-capita healthcare employment, while access works to raise it. We organize our hypothesis around the property-rights theory of the firm (Hart et al., 1997; Shleifer, 1998), which states that when quality cannot be fully specified in contracts, ownership determines how strong the incentive to cut costs is. Under private ownership the provider keeps what it saves and cannot be penalized for cutting quality on dimensions no contract covers, so its incentive to economize increases while its incentive to sustain noncontractible quality decreases, and because labor is the largest variable input in care delivery, this falls first on staffing, and the ownership margin predicts $\Delta L^{\text{ownership}} \leq 0$. Capitation creates the same cost-cutting incentive. Given that a fixed amount per member is paid, a provider becomes the residual claimant on what it does not spend and so has a direct reason to decrease the inputs it controls, including labor (Aizer et al., 2007; Duggan, 2004; Marton et al., 2014). Capitation therefore also predicts fewer healthcare workers, $\Delta L^{\text{capitation}} \leq 0$, unless the capitated rate rises to offset it.

The implication for per-patient health, which comes as a consequence of noncontractible quality q, is two-sided and depends on the level from which the reform started. The same property-rights logic admits a gain as well as a loss: Hart et al. (1997) hold that private provision is generally cheaper but may raise or lower quality, with the sign depending on how much non-contractible quality a cost cut sacrifices. Where the public system had been adequately resourced, that cost cutting lowers quality, $\Delta q \leq 0$. Where it had instead been underinvesting, and where the binding margin is physical capacity rather than the non-contractible effort the model emphasizes, transferring delivery to better-equipped private providers can raise quality, as competition does in public hospital systems at administered prices (Propper, 2018). Puerto Rico's pre-reform system was the underinvesting case (according to (Commonwealth of Puerto Rico, 1993)), so the sign of Δq is an empirical question whose answer also depends on how we value the inputs the reform removed, a point we return to in the welfare analysis (Section 6.4).

The access margin is ambiguous. Extending private insurance coverage to the uninsured (assuming no take-up friction) raises the demand for care, which should expand the local healthcare labor market (Dillender, 2022; Finkelstein, 2007; Hackmann et al., 2025). Routing those enrollees through a single managed-care network, with gatekeeping and cost-sharing, works the other way by decreasing utilization. The net sign of ΔL^{access} is therefore an empirical question. Summing them gives the reform's net effect,

$$\Delta L^{\text{reform}} = \underbrace{\Delta L^{\text{ownership}}}_{\leq 0} + \underbrace{\Delta L^{\text{capitation}}}_{\leq 0} + \underbrace{\Delta L^{\text{access}}}_{\gtreqless 0}. \tag{1}$$

Our estimate represents the combined effect, not the contribution of each, because the margins interact. For example, the incentive to economize under capitation is stronger when the provider also owns the facility. The access margin interacts with the other two as well when the coverage expansion raises demand, but whether that demand reaches the low-income patients depends on whether private, capitated providers serve the unprofitable patients the expansion was meant to cover rather than selecting toward more lucrative ones, so coverage and the supply-side incentives act as complements. In the single-hospital privatizations of Duggan et al. (2023), low-income admissions fell at the market level even with coverage unchanged, the supply-side response our access margin would have to overcome.

These three margins dominate in different counties. This change in access may have a bigger effect in counties where coverage gains were largest (poorer, higher-unemployment counties). Yet, we may have seen a contraction in counties where the privatized public sector was largest (denser ones). We read this cross-sectional contrast as suggestive only, since county types differ along the same urban gradient that set the rollout order and the subsamples are too small to test formally; it cannot attribute the effect to a single margin, though the concentration of the contraction in dense, high-capacity markets is itself a feature of the estimates we report[9].

[9] Appendix 10 reports the formal derivations of the ownership and capitation predictions, the quality corollary, and the mobility margin.

5 Empirical Strategy

The empirical analysis exploits the staggered rollout of the reform across five cohorts of counties ($\mathcal{G} = \{1994, 1995, 1996, 1998, 2000\}$) to identify the average effect of treatment on the treated (ATT). Let $Y_{i,t}$ denote an outcome measured for county i in year t, let $G_i \in \mathcal{G}$ denote the treatment cohort of county i, and let $D_{i,t} = \mathbf{1}\{t \geq G_i\}$ be the post-treatment indicator. Because every county in Puerto Rico is treated by 2000, the analysis has no never-treated comparison group. Therefore, identification relies instead on *not-yet-treated* counties, units in later cohorts $g' > g$ not yet treated by year t. The estimand of interest is the group-time average treatment effect $\text{ATT}(g,t) = \mathbb{E}[\,Y_{i,t}(g) - Y_{i,t}(\infty) \mid G_i = g\,]$ for $t \geq g$ (Callaway & Sant'Anna, 2021), where $Y_{i,t}(g)$ and $Y_{i,t}(\infty)$ are the potential outcomes under treatment-in-cohort-g and no-treatment respectively. Identification rests on three assumptions: (i) conditional pre-treatment parallel trends for the not-yet-treated comparison group, $\mathbb{E}[Y_{i,t}(\infty) - Y_{i,t-1}(\infty) \mid G_i = g] = \mathbb{E}[Y_{i,t}(\infty) - Y_{i,t-1}(\infty) \mid G_i > t]$; (ii) no anticipation, $Y_{i,t}(g) = Y_{i,t}(\infty)$ for $t < g$; and (iii) staggered, irreversible adoption. Under (i)–(iii), $\text{ATT}(g,t)$ is identified by the doubly robust difference-in-differences estimator

$$\widehat{\text{ATT}}(g,t) \;=\; \mathbb{E}\big[Y_{i,t} - Y_{i,g-1} \,\big|\, G_i = g\big] \;-\; \mathbb{E}\big[Y_{i,t} - Y_{i,g-1} \,\big|\, G_i > t\big]\,. \tag{2}$$

The regression tables report the simple weighted average of $\widehat{\text{ATT}}(g,t)$ across cohorts and post-treatment periods, with weights proportional to the number of post-treatment county-year observations; the event-study figures report the dynamic aggregation $\widehat{\theta}_\ell$ for $\ell \in \{-4, \ldots, +4\}$ with the reference period $\ell = -1$. Inference uses the multiplier bootstrap of Callaway and Sant'Anna (2021) clustered at the county level with 1,000 replications.

Because the estimand and the parallel-trends assumption are estimator-specific, Section 7 re-estimates every main outcome with three alternatives: the canonical OLS two-way fixed-effects estimator (TWFE), the extended two-way fixed-effects (ETWFE) estimator of Wooldridge (2025), and the imputation estimator of Borusyak, Jaravel, and Spiess (2024)

(BJS). Additionally, since earlier-treated counties are systematically more rural than later-treated ones, the rollout produces cohort imbalance on observable pre-treatment levels (as seen in Table 2), but under Callaway and Sant'Anna (2021) (c&S) such imbalance does not threaten identification so long as the not-yet-treated comparison group satisfies assumption (i). We assess that assumption directly through the event-study pre-trend coefficients reported alongside each main result in Section 6.

Three features of the setting depend on the identifying assumptions, and Section 7 tests each of them. The first is geography, and it works in the design's favor. As an island, we assume that Puerto Rico generated no spillovers to healthcare markets beyond it, removing the cross-border confound that complicates privatization studies that exploit policies within the continental USA. The residual concern is internal, in labor and patient flows across treated and not-yet-treated counties that would contaminate the comparison group assumption (i) relies on; we assess it by re-estimating on counties that share no border with a county treated in a different year. The second is the IRS Section 936 corporate-tax phase-out (a generous corporate tax incentive), which began in 1996 and is contemporaneous with the rollout; because it fell unevenly across counties by manufacturing exposure, it is a potential confound, and we condition on manufacturing employment to test whether it accounts for the estimates. The third is the leverage of San Juan as the sole 2000 cohort. With no county treated later, its group-time effect has no not-yet-treated comparison and is weakly identified, so we report its influence separately.

6 Results

6.1 First Stage and Supply Contraction

Table 3 and its event study Figure 1 panel (b) report the estimates on the share of births delivered at government and at private hospitals. This is the most available proxy we have in our data for which to measure how ownership affected the inpatient delivery system.

The government-hospital share fell by 14.79 percentage points ($p < 0.01$), or 23.6 percent of its 62.55 percent pre-reform mean, and the private-hospital share rose by an offsetting 15.19 percentage points ($p < 0.01$). The two estimates are statistically indistinguishable in absolute magnitude and opposite in sign. This is because the reorganization of births is essentially a one-for-one substitution from public to private facilities.[10] Additionally, in the event study we can confirm that the timing of the divergence in trends coincides with the start of the policy.

Coincident with the ownership transition, per-capita healthcare employment fell. Table 4 reports the labor-market estimates for the healthcare aggregate (NAICS 621+622+623) and, separately, for ambulatory primary care (NAICS 621), the subsector most directly associated to the privatized county clinic network. Healthcare employment per thousand residents fell by 0.511 employees, 13.0 percent of its pre-treatment mean ($p < 0.01$), with a larger decline of 17.3 percent in ambulatory primary care ($p < 0.01$). Establishments per thousand fell by 6.6 percent in the aggregate ($p < 0.01$) and by 4.6 percent in NAICS 621 ($p < 0.10$), so the contraction operated on both the extensive margin (net establishment exit) and the intensive margin (fewer employees per establishment).

This per-capita decline is a population-weighted average, and it does not mean the sector shrank everywhere. In a $\log(1 + x)$ specification the same outcomes diverge in sign from the per-capita levels (see Appendix Table A.7): the per-capita measure is outweighed by the population dense urban counties, where the sector contracted in absolute terms, while the log measure reflects proportional expansion among the smaller rural counties. The policy therefore affected healthcare activity more in the dense urban centers, where healthcare markets are more concentrated, than its rural counterparts. This finding aligns with Duggan et al. (2023), which documents that privatization effects are more severe in highly concentrated markets.

The main effect was seen on the quantity of labor, not its price. According to Table 4

[10] The two shares are not exact complements because a small fraction of births, typically under one percent, is recorded as non-hospital or with unspecified ownership.

employment and establishment counts fell while compensation per worker stayed flat, with the wage-per-employee estimate statistically indistinguishable from zero in both panels. This confirms our hypothesis in Section 4, where the ownership and capitation mechanisms incentivize providers to reduce their largest variable input rather than cut wages. This pattern is also documented across the literature, on both the delivery and the payer side. For example, Duggan et al. (2023) find that privatized U.S. hospitals cut labor intensity by about 8 percent with no significant change in compensation, and the same input economizing appears under Medicaid capitation (Duggan, 2004). On the payer side, estimates indicate that Medicare Advantage spends 9 to 30 percent less per enrollee than traditional Medicare through lower utilization rather than lower prices (Curto et al., 2019), and privatizing the Medicaid drug benefit cut spending by about 21 percent through substitution toward cheaper products (Dranove et al., 2021). What sets our result apart is its locus, where those studies describe a private insurer economizing on the care it pays for, ours describes the delivery system itself shrinking the inputs it operates.

Figure 2 reports the corresponding six-panel event study, for employment, establishments, and the wage bill per employee across both industry definitions. The pre-treatment coefficients are statistically flat through $\ell = -1$, satisfying the parallel-trends assumption, and the post-reform contraction in employment and establishments starts within the first year while the wage-per-employee stays flat throughout.

The establishment decline is also interpreted as a change in market structure. Fewer establishments per resident means fewer providers competing in each market, most of all in urban areas where the contraction concentrated. A literature on hospital competition finds that this matters for care. Its been found that in the English NHS, where hospitals compete on quality at administered prices, greater competition has been linked to lower mortality and better management (Bloom, Propper, Seiler, & Van Reenen, 2015; Cooper, Gibbons, Jones, & McGuire, 2011; Gaynor, Moreno-Serra, & Propper, 2013). Reduced competition therefore gives the reform a second route to quality, separate from the cut in inputs, and one a single-facility privatization cannot identify. Thus, the net effect on quality is ambiguous. The loss of competition may push it down, while the replacement of an underinvesting public monopoly with private provision may push it up, and we take up the realized effect with the patient-health estimates in Section 6.3.

Physical hospital capacity contracted as well, on the narrower CMS panel of forty one Medicare-certified counties. These estimates appear alongside the patient-health outcomes in Table 5, Columns 3–4: Medicare-certified hospital beds per thousand fell by 7.8 percent ($p < 0.01$) and Medicare-certified services per thousand by 4.9 percent ($p < 0.05$). The contraction therefore spans both layers of the delivery system, the ambulatory clinic network (QCEW employment and establishments) and the inpatient hospital infrastructure (CMS beds and services). The bed and service event studies appear in Figure 3, Panels (e)–(f), where the post-treatment contraction opens by $\ell = +2$ [11].

6.2 Worker-Side Adjustment Margins

The labor-market contraction did not show up in wages or in the unemployment rate. Table 6 reports the margins along which displaced healthcare workers and broader local-labor-market spillovers might adjust. The labor-force-participation rate fell by 1.11 percentage points, 3.5 percent of its 31.895 percent pre-treatment mean ($p < 0.01$), while the BLS-standard unemployment rate is statistically indistinguishable from zero against a pre-reform mean of 17.770 percent [12]. The crude net migration rate moves from a pre-treatment mean of +1.06 residents per thousand to an estimated post-treatment −14.65 per thousand ($p < 0.05$), and the crude birth rate falls 3.1 percent (−0.564 per thousand, $p < 0.05$). Total county population, by contrast, doesn't decrease. The population-level estimate is +5.1 percent ($p <$

[11] The supply magnitudes reported here are simple-aggregate ATTs, averaged over all post-treatment periods. The event studies in Figures 2 and 3 show the per-capita contraction deepening with time since treatment, so the longest-horizon coefficients exceed these averages, with correspondingly wide confidence intervals. We report the simple-aggregate ATT and use it in the welfare accounting (Section 6.4) as the conservative choice.

[12] The unemployment rate is reported in Online Appendix Table OA.1, alongside the labor-force-participation and employment-to-population rates; Appendix 10 reports the participation event study, who satisfies parallel trends and starts declining as of the policy rollout.

0.01).[13] These estimates rule out a population exodus as the driver of the contraction, since treated counties grew even as participation and the crude birth rate fell. We do not observe age composition and so infer no specific demographic recomposition. The participation and fertility declined, but not as an effect of population contraction. The net-migration and crude-birth-rate event studies appear in Figure 3, Panels (c)–(d), the migration trend changing within two years of treatment and the birth-rate trajectory bending downward from $\ell = 0$.

These margins also bound what we can say about the displaced workers themselves. Because unemployment did not rise, they were not absorbed into local unemployment, since the effect was negative for labor force participation and migration. County aggregates cannot follow individuals, so we cannot separate labor-force exit, migration off the island, and reallocation into other local sectors. Two features make selective provider emigration the more plausible hypothesis. The population grew, so any migration was selective rather than broad, and Puerto Rico's clinicians hold U.S. credentials that require no re-licensing on the mainland (Appendix 10), consequently lowering the cost of leaving for precisely the workers the reform displaced. The negative net-migration estimate is consistent with this selective outflow.

The net migration rate is the one primary outcome whose sign is sensitive to estimator choice, so we read its magnitude as suggestive (Section 7). The participation result, on the same 78-county panel and the behavioral counterpart of the migration hypothesis, is robust across all four estimators.

[13]Online Appendix Tables OA.20 and OA.21 report the population estimate in levels and logs and an explicit demographic-balancing check. The population-level and net-migration estimates do not arithmetically reconcile, because the migration rate is a residual of three separately measured flows (see Equation 4), more sensitive to compounding measurement error than the level outcome, and because the Callaway and Sant'Anna (2021) aggregation weights differ across panels of different length.

6.3 Patient Health, Heterogeneity, and Access

The short-horizon effect on patient health is null on average. Table 5, Columns 1–2, reports the county-aggregate Vital Statistics outcomes whose response horizons fall within the four-year post-reform window: the infant mortality rate fell by 1.205 deaths per thousand live births (9.5 percent of its pre-mean, statistically insignificant) and the low-birth-weight share rose by 0.06 percentage points (statistically insignificant), and both event studies in Figure 3, Panels (a)–(b), are flat in the pre- and post-treatment windows.[14]

The cause-specific mortality estimates in Appendix Table A.9 refine this average null into a set of offsetting movements across two groups that proxy different dimensions of care. The first collects deaths that timely preventive and chronic-disease care can avert, and on the cardiovascular margins it points to improvement. Heart-disease mortality falls 6.1 percent ($p < 0.05$) and cerebrovascular mortality 11.0 percent ($p < 0.05$), while diabetes and pneumonia or influenza mortality are unchanged. Maternal mortality moves the other way, rising 73 percent of its small pre-reform base (+0.002 deaths per thousand residents, $p < 0.10$); because a maternal death usually reflects prenatal risk left unmanaged rather than delivery-room failure, this is consistent with acute delivery capacity holding while the preventive care that depends on sustained patient contact decreased. The maternal margin is the population-health counterpart of the selective retrenchment Duggan et al. (2023) document, in which private providers withdraw from the least profitable, low-income service lines. The second group, deaths from accidents and homicide, proxies emergency and trauma care, where survival turns on acute-care access that often does not cross county lines (since time is of essence); both rise, accident mortality by 40 percent (but statistically insignificant) and homicide mortality by 75 percent ($p < 0.01$), consistent with the contraction of beds and facilities lengthening the path to emergency care. The reform left little capacity elsewhere to absorb the displaced demand, since each region was served by a single managed-care network,

[14] Individual-level NVSS specifications on birth weight, prenatal-visit count, trimester of prenatal-care initiation, and Apgar scores are similarly null (see Online Appendix Table OA.22).

the limiting case of the concentrated-market condition under which Duggan et al. (2023) find competing hospitals do not take up the slack and market-level low-income admissions fall. We read the homicide estimate cautiously, because its recorded count blends the incidence of violence, which rose across Puerto Rico through the 1990s for reasons unrelated to the health system. These acute-care deaths are readable within the four-year window through case-fatality, where chronic-disease mortality is not, and the Hispanic health paradox (Fernández, García-Pérez, & Orozco-Alemán, 2023) implies measured health can deteriorate well before mortality responds, even if this population seems to have lower mortality rates. Across the within-window outcomes, then, the privatized system delivered fewer per-capita inputs with offsetting movements in health rather than a broad deterioration. Taken together, the cause-specific decomposition suggests an improvement on the cardiovascular margins, no average change in infant mortality, and deterioration on the obstetric margin and, more tentatively, on emergency-care access.

Table 7 partitions the 78 counties at the median pre-reform government-hospital birth share, and the first-stage substitution is larger in the below-median group (−15.49 percentage points, $p < 0.01$) than the above-median group (+2.41 percentage points, statistically insignificant). The baseline share is the wrong intuition on where the reform bit, because pre-reform reliance on government facilities was highest in the rural, poorer counties with the fewest count of healthcare facilities. That high-reliance half is largely the rural periphery, where the same facilities went on delivering most births, so measured substitution was small. The realized ownership transition instead concentrated in the urban, more populous, higher-capacity, richer, and lower-unemployment counties. Partitioning on pre-reform population density, population, market concentration (healthcare employment), the poverty rate, or the unemployment rate, the government-hospital share falls by 24 to 31 percent in the urban, larger, higher-capacity, richer, and lower-unemployment half and is flat in the other (see Appendix Table A.6 and Online Appendix Tables OA.9, OA.10, OA.23, and OA.24).

The infant-mortality rate follows the same pattern. In the halves where the ownership transition took hold, infant mortality falls by 13 to 27 percent of its pre-treatment mean, statistically significant across the density, population, market concentration (healthcare-employment), poverty, unemployment, and commuting partitions; in the halves where the substitution was muted it does not improve, and in the smallest counties it rises, a marginally significant 28 percent.[15] The quality gain follows the realized ownership transition, concentrated in the urban counties with the private infrastructure to absorb the transfer, the lower-poverty, higher-capacity markets that Duggan et al. (2023) find most resilient to an ownership change, consistent with the gains the theory predicts for under-resourced public systems. These counties differ on many dimensions at once and the subsamples are small, so we read the pattern as consistent with quality gains from the transition rather than a clean isolation of which margin produced them. The improvement does not concentrate in the poorest counties, where the coverage expansion delivered its largest gains, so it does not trace the access margin, and it is hard to align with the uniform quality deterioration Hart et al. (1997) predicts, though it falls within the boundary condition of that theory set out in Section 4 rather than refuting it: an under-resourced public baseline with physical capacity, not non-contractible effort, as the binding margin. It also runs opposite to Duggan et al. (2023), who find that U.S. hospital privatization raised elderly mortality. The difference is largely based on population and time horizon, since their effects fall on a horizon our four-year perinatal window does not reach, while among the adult causes we can read at a short horizon, maternal mortality moves up as their elderly mortality does and cardiovascular mortality moves down. Unlike the payer-only managed care that Aizer et al. (2007) find raised neonatal mortality, the reform bundled a coverage expansion with delivery privatization, and the infant gains concentrate where private infrastructure absorbed the transfer. The

[15] For the half in which the ownership transition concentrated, the infant-mortality ATTs in deaths per thousand live births are -2.03 (urban, $p < 0.05$), -2.56 (larger population, $p < 0.01$), -2.63 (higher healthcare employment, $p < 0.01$), -1.62 (lower poverty, $p < 0.10$), -2.62 (lower unemployment, $p < 0.05$), and -3.12 (workplace-center, $p < 0.01$); the government-hospital substitution in these halves ranges from -14.2 to -18.2 percentage points. In the opposite halves the infant-mortality estimates are positive and insignificant, except the smallest-population half at $+3.66$ ($p < 0.10$). Appendix Table A.6 and Online Appendix Tables OA.9, OA.10, OA.23, OA.24, and OA.11 report all six partitions.

payer-side evidence fits the no-harm side, since Duggan et al. (2018) find that Medicare Advantage restricted hospital access with no measurable mortality effect, comparing a private payer to a public one rather than private to public delivery, so it bounds our interpretation. Because mortality is a coarse proxy for quality, a caveat Duggan et al. (2018) attach to their own no-harm finding, our null cannot rule out movements in dimensions of care that mortality does not capture. A measurement caveat from the same literature does not bind here: Chorniy et al. (2018) show that managed-care risk-adjustment and quality-reporting incentives inflate recorded diagnosis and treatment counts, a sensitivity our physical-capacity and vital-statistics outcomes do not share. The supporting heterogeneity event studies, by baseline poverty rate and by pre-reform government-hospital exposure, appear in Figures 4 and 5, each splitting the panel at the median and tracing employment, infant mortality, and net migration for the two halves.

The access margin moved in the expansionary direction. Across age, education, and employment subgroups, self-reported insurance coverage in the BRFSS rises by 3 to 10 percentage points, estimated under the Wooldridge (2025) extended two-way fixed-effects estimator because the post-1996 BRFSS window admits only the later cohorts.[16] An administrative series corroborates this survey-based result. The Medicaid managed care enrollment introduced in Section 2.2 tracks the cohort rollout (Appendix Figure A.3) and is the program-level counterpart to the survey-measured coverage gain (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1999).[17] This is the opposite of what facility-level privatization produces in the United States, where Duggan et al. (2023) find market-level admissions of low-income patients falling. The Puerto Rico reform was instead built as a coverage expansion for the low-income and uninsured, so its enrollees were the

[16]Online Appendix Tables OA.14–OA.19 report the six BRFSS subgroup estimates; the "has health plan" point estimates range from +2.8 percentage points (ages 65 and older, where near-universal Medicare leaves little room to expand) to +10.5 percentage points (ages 45–64), and are significant in five of the six subgroups. Appendix 10 discusses the subgroup pattern and the single-cohort-exposure limitation that requires the ETWFE estimator on this panel.

[17]Enrollment is read from the published budget figure and is approximate; it corroborates the access margin descriptively and does not identify a separate effect.

low-income group itself, and access on the insurance margin widened rather than narrowed. This coverage expansion is the access margin's empirical counterpart; combined with the supply contraction documented above, it shows the contractionary equilibrium held even as coverage expanded, an aspect in which the supply-stimulus prediction of Finkelstein (2007), Dillender (2022), and Hackmann et al. (2025) reverses sign once coverage is bundled with delivery-system privatization, capitation, and a single managed-care network.

6.4 Welfare Bounds

We close by combining the estimated supply and health effects into a back-of-the-envelope cost-benefit calculation, following Duggan et al. (2023). We report it in constant 2025 dollars throughout, as elsewhere in the paper.[18] Read for sign and order of magnitude rather than as a point estimate, it treats the beds and workers the reform freed as a welfare gain under two assumptions: that care quality is maintained, which the data support only as the absence of a detectable full-panel change, and that the saving reflects freed real resources rather than reduced provider rents.

We define net welfare as the avoided cost of the healthcare inputs the reform no longer purchased plus the value of any averted infant deaths:

$$W = \underbrace{\frac{P\,T}{1000}\Big(|\hat{\beta}_{\text{bed}}|\,c_b + |\hat{\beta}_{\text{emp}}|\,\bar{w}\Big)}_{\text{avoided cost}} + \underbrace{v\,\frac{|\hat{\beta}_{\text{IMR}}|}{1000}\,B_H\,T}_{\text{value of averted infant deaths}} . \tag{3}$$

We depart from Duggan et al. (2023) in three respects. First, the cost saving is the avoided cost of the inputs the delivery system saved, the per-thousand reductions of $\hat{\beta}_{\text{bed}} = -0.265$ beds (see Table 5) and $\hat{\beta}_{\text{emp}} = -0.511$ workers (see Table 4)[19] scaled to the island population

[18] The calculation is carried out in real 1982–1984 dollars to match the price base of Duggan et al. (2023) and converted to constant 2025 dollars with the U.S. CPI-U (all items, U.S. city average, 1982–1984 = 100; 2025 annual average ≈ 321.9, a factor of about 3.22). The 1982–1984-dollar values are reported in the footnotes that follow and in Appendix Table A.4, which gives every cell in both price bases.

[19] These are the simple average treatment effects on the treated, not the larger end-of-horizon event-study coefficients; pricing the period-average rather than the peak effect keeps the avoided-cost term conservative.

over the four post-reform years; we measure the physical inputs directly, since the health reform privatized an entire delivery system whose budget we do not observe at the provider level.[20] A bed-year is priced at its implied total operating cost c_b from the CMS cost reports (Centers for Medicare and Medicaid Services, 2023a) and a worker-year at the 1990–1993 mean healthcare wage $\bar{w}$. Second, the health term is a benefit rather than a cost, since infant mortality fell where they find added deaths among the elderly. Third, because the full-panel infant-mortality reduction is statistically insignificant, we value it only as an upper bound, applying it over the births in the high-substitution half B_H of Section 6.3, one-half the island total, at the U.S. EPA reference value of \$10 million per statistical life (U.S. Environmental Protection Agency, 2010; Viscusi & Aldy, 2003).[21]

Under the zero-health choice the bound rests on the avoided cost alone and is \$967 million under the employment estimate (\$69 per resident per year). Conditioning the employment reduction on the contemporaneous manufacturing-tax phase-out (Section 7.5) attenuates it to -0.224 per thousand and lowers the avoided cost to \$844 million (\$60 per resident per year), our most conservative figure. Adding the value of the averted infant deaths, about \$3.1 billion, raises net welfare to \$4.0 billion under the main supply estimate and \$3.9 billion under the §936-conditional one (\$289 and \$280 per resident per year).[22]

[20] This avoided cost is the value of the physical inputs the system no longer used, not a reduction in public outlay. Aggregate public health spending in fact rose over this period, as purchasing through ASES scaled past \$1 billion in nominal terms (Appendix 10), so the welfare improvement is a fall in real-input use under maintained quality rather than a smaller public budget, a pattern consistent with Duggan (2004), who finds that mandatory Medicaid managed care raised government spending.

[21] This \$10 million reference value is in 2024 dollars, equivalent to about \$10.3 million in the constant 2025 dollars used throughout the paper and about \$3.19 million in the 1982–1984 dollars in which the calculation is carried out.

[22] In 1982–1984 dollars these are an avoided cost of \$300.8 million (main supply estimate) and \$262.5 million (§936-conditional), an averted-infant-death benefit of \$958.7 million valued at a \$3.19 million statistical life, and net welfare of \$1,259.5 million (main) and \$1,221.2 million (§936-conditional); Appendix Table A.4 reports every cell in both price bases. This upper bound values only the averted infant deaths, and on a footing favorable to the reform: the infant benefit is recovered in the high-substitution subgroup, while on the full panel the infant-mortality effect is statistically indistinguishable from zero. The contemporaneous maternal-mortality increase (see Appendix Table A.9) is a full-panel estimate working the other way; we flag it as an offset rather than netting it in, because the marginally significant maternal coefficient off a very small base is not on the subgroup footing used for the infant term. The asymmetry of footings is itself why we read these health-valued cells as an upper bound: on a single consistent footing the monetized health term would be much smaller, leaving the avoided cost as the binding source of the welfare gain.

We take the zero-health alternative as our main estimate, since they are positive, they rest only on the avoided cost, and do not depend on the patient-health margin in either direction; among them we prefer the §936-conditional figure, which nets out the manufacturing-tax phase-out. The health-valued cells are an upper bound, for three reasons. The infant-mortality benefit they monetize is insignificant on the full panel and recovered only in subgroups; they price that benefit while ignoring the adult cause-specific effects, which move both ways, the favorable cardiovascular declines against the adverse maternal increase, the clearest identified short-horizon harm; and they apply an adult statistical-life value to infant deaths, understating the per-death value, since an averted infant death returns far more life-years than the elderly deaths Duggan et al. (2023) price, so valuing the health term in life-years rather than in lives would only raise our benefit. We clarify that there are four additional margins we cannot quantify, but none of them changes the sign of our estimate: mortality beyond the four-year window, the value of the coverage expansion to enrollees (which low-income willingness-to-pay places below its cost (Finkelstein, Hendren, & Shepard, 2019)), the share of the saving that reflects reduced provider rents rather than freed real resources, since much healthcare spending is such a transfer (Finkelstein, Hendren, & Luttmer, 2019), and whether the freed labor and capacity were redeployed productively, since the welfare value of freed inputs depends on frictions in the input market (Hackmann et al., 2025). Read in these terms, our result stands opposite the single-hospital case. The same accounting leads Duggan et al. (2023) to judge that case socially inefficient, since the roughly \$2.7 million their average privatization saves the local government comes with about 3.6 additional elderly deaths, some \$0.8 million of saving per death lost and an order of magnitude below the \$10 million value of a statistical life. Our positive sign instead rests on the inputs the system saved standing against a health margin that improved on its dominant component, so the bound clears zero even before any health benefit is counted.

7 Robustness

7.1 Estimator class

We compare the main estimates against three alternatives: the canonical two-way fixed-effects estimator, the extended two-way fixed-effects estimator of Wooldridge (2025), and the imputation estimator of Borusyak et al. (2024)[23]. We rely on TWFE to understand the source of our bias. That is, if the treatment effects vary across cohorts and time (which it does, since we move from rural to more urban counties as the policy rolls out), its coefficient is a negatively weighted average that can bias the estimator. To account for this bias, we prefer to rely on the C&S, ETWFE, and BJS estimators, since they are designed to avoid this. This is why ETWFE, C&S, and BJS agree in sign and magnitude on the first-stage transition, the healthcare-employment contraction, the labor-force-participation decline, and the hospital-bed contraction, while TWFE produces the expected sign reversals on the outcomes where heterogeneity is most pronounced (healthcare employment, infant mortality, and the low-birth-weight share).[24] The crude net migration rate is the one outcome on which the four estimators disagree. TWFE, ETWFE, and BJS return insignificant estimates of mixed signs (-5.69, -1.53, and $+4.51$ per thousand), whereas the C&S estimator yields a significant -14.65 per thousand ($p < 0.05$), the estimate on which the migration discussion in Section 6.2 rests. We therefore read it as suggestive of net out-migration, relying on its sign rather than its magnitude as a result.

[23](Refer to Appendix Figure A.1 for the comparison event studies across all estimators)

[24]Appendix Table A.1 reports the four-estimator point estimates and bootstrap standard errors for the six primary outcomes, and the full TWFE specification (county and year fixed effects with event-time indicators, reference period omitted, standard errors clustered at the county level). Across estimators, the first-stage government-hospital-share estimate is -12.9 percent (TWFE), -26.4 percent (ETWFE), -23.6 percent (C&S), and -18.0 percent (BJS) of pre-mean, all $p < 0.01$; HC employment per thousand is -25.7 (ETWFE), -13.0 (C&S), and -16.7 (BJS), all $p < 0.01$; LFP per population is -3.6, -3.5, and -2.2 percent, all $p < 0.01$; hospital beds per thousand is -13.1, -7.8, and -5.6 percent, all $p < 0.01$.

7.2 Excluding San Juan

San Juan is the sole 2000-cohort county and holds roughly 11 percent of the Commonwealth's population. Dropping it attenuates the supply-side and labor-market estimates while leaving the first stage intact. When doing so, the government-hospital-share estimate moves from -14.79 to -8.22 percentage points (both $p < 0.01$),[25] healthcare employment from -13.0 to -5.3 percent of pre-mean and the hospital-bed estimate from -7.8 to $+3.0$ percent, the latter two no longer significant. The attenuation is mechanical, since the capital holds the largest concentration of healthcare workforce and Medicare-certified capacity, so dropping it strips out most of the units of observation through which a per-capita contraction is identified. Where that concentration sits is itself informative. The capital is not only the largest market but the lowest-poverty and highest-capacity one, the setting in which Duggan et al. (2023) find privatization least likely to reduce access, because the surrounding market can absorb the change. The reform's largest input reduction therefore fell where access was most insulated, and the infant-mortality improvement concentrates in these same counties (Section 6.3); the San Juan-inclusive magnitude is the effect where the public system was largest and most readily absorbed by the private sector.

7.3 Within-island spillovers

The not-yet-treated comparison requires that already-treated neighbors not affect comparison counties (Section 5). We rely on two restrictions to understand this effect: a non-border subset (where we drop counties that share a boundary with a different-cohort treated unit), and a low-commuting-inflow subset (where we drop the upper half of the workplace in-commuting distribution (U.S. Census Bureau, 2000a)).[26] The first-stage transition is

[25] Online Appendix Table OA.3 reports the first-stage and patient-health side-by-side specifications; Appendix Table A.2 reports the QCEW labor-market and CMS hospital-capacity specifications; Online Appendix Table OA.4 reports the ownership-stratified CMS robustness.

[26] Appendix Table A.5 and Online Appendix Tables OA.6, OA.5, OA.7, and OA.8 report the corresponding specifications. The non-border restriction retains 27 of 78 counties on the QCEW panel, 19 of 40 on the CMS panel, and 39 of 78 on the Vital Statistics panel. The low-commuting-inflow restriction retains 64 of 78 on the Vital Statistics panel and 34 of 40 on the CMS panel.

robust under both (-12.14 percentage points on the non-border subset, -7.18 on the low-commuting subset, both $p < 0.05$), as are the patient-health outcomes. The labor-market estimates do not transfer to the non-border restriction at its sample size of 27 QCEW counties, where healthcare employment, the wage bill per employee, and establishments are each indistinguishable from zero, and the hospital-bed and services estimates keep the sign of our previously estimated coefficient but lose precision. We read this as an effect of power loss rather than evidence against the main estimates.

7.4 Placebo industries

These specifications test whether contemporaneous shocks are orthogonal to the reform (for example, the Section 936 corporate-tax phase-out beginning in 1996), which we test to see if it produced healthcare effects on industries with no exposure to the privatized delivery system: NAICS 11 (agriculture), NAICS 48–49 (transportation), and NAICS 71 (arts). NAICS 31–33 manufacturing is excluded *a priori*, since the IRS Section 936 phase-out shocks it during the window, making it endogenous to the same policy as healthcare rather than orthogonal.[27] The evidence is mixed. Agriculture is null on absolute-headcount and per-capita employment and establishments, its per-capita wage-bill decline not surviving raw-level normalization (which could be a population redistribution, and not an agriculture-specific shock). Transportation contracts in both per-capita and raw-levels beyond what redistribution can produce, consistent with input-output linkage to the phasing-out manufacturing supply chain through manufacturing sub-industries. Arts is underpowered ($n = 9$). The transportation failure through a known confounder motivates the direct manufacturing-control test specified next.

[27] Online Appendix Tables OA.12 and OA.13 report the per-thousand-resident and raw-level placebo specifications.

7.5 Manufacturing control

The transportation placebo failure motivates a direct test of whether the main contraction is identified in isolation from the IRS Section 936 phase-out. We condition the main healthcare-employment estimate on a pre-reform manufacturing-exposure covariate, the 1990–1993 mean of NAICS 31–33 employment per thousand; using a strictly pre-1996 baseline rather than the contemporaneous series ensures the covariate is not itself shocked by the policy whose contamination the test nets out.[28] The C&S estimate attenuates from -14.7 percent of pre-mean (no control, $p < 0.01$) to -6.8 percent (with the covariate, statistically insignificant), remaining negative and not driven by sample restriction. The result is therefore neither pure §936 contamination, since the conditional estimate is negative rather than zero, nor identified in isolation at full magnitude. We interpret this as part of the cross-sectional heterogeneity in the contraction being reflected in the pre-reform manufacturing exposure, a correlation that may reflect §936 contamination but may also reflect that high-manufacturing counties are urban and industrial and so overlap with the dimensions on which the healthcare contraction concentrates (Section 6.3).

7.6 Per-capita normalization

Our data shows a -13.0 percent contraction in healthcare employment per thousand and a roughly $+19.0$ percent growth in raw-level $\log(1 + y)$ employment. We interpret this as the local healthcare sector growing in absolute terms but more slowly than the local population.[29] Therefore, we interpret this difference as distributional. The $\log(1 + y)$ raw-level measure weights counties roughly equally and is pulled upward by the many low-baseline rural counties that expanded proportionally, whereas the per-1,000 measure is population-weighted and

[28] Appendix Table A.3 reports the conditioning specification. The sample is the 54 counties with non-missing 1990–1993 baseline manufacturing data. The C&S no-control estimate on this sample is -14.7 percent of pre-mean, economically equivalent to the 78-county main result of -13.0 percent reported in Table 4, so the sample restriction itself does not drive the conditioning result.

[29] Appendix Table A.7 reports per-1,000 and $\log(1 + y)$ specifications for QCEW healthcare employment, establishments, and the wage bill side by side; Online Appendix Table OA.2 reports the parallel specification on $\log(1 + y)$ per-1,000 normalizations and $\log(1 + y)$ raw-level outcomes.

dominated by the high-baseline urban centers, which contracted both absolutely and per resident. For example, the raw-level estimate is the change in the absolute job count, the per-1,000 estimate the change in jobs per resident, and the $\log(1+y)$ estimate the change in proportional terms. The +19.0 percent is the $\log(1+y)$ figure, driven up by the small, low-baseline rural counties where adding a few jobs is a large percentage gain, whereas the −13.0 percent is the per-1,000 figure, driven down by the large urban centers, where most residents live and where jobs per resident fell.

8 Conclusion

Puerto Rico's 1993 health reform sold or leased its public healthcare facilities to private providers. At the same time, it also shifted to paying private managed-care organizations prospective capitation and changing its coverage policies. We find that this policy decreased healthcare resources per capita: healthcare employment, ambulatory primary-care employment, the establishment count, and Medicare-certified hospital beds and services all contracted relative to population. The contraction was seen in the labor-force participation rather than unemployment, alongside declines in the crude birth rate and net migration; the displaced workers were not absorbed into local unemployment, and the participation and migration declines are consistent with labor-force exit. Yet, patient health did not broadly deteriorate within the four-year window and improved where the ownership transition took hold, infant mortality falling 13 to 27 percent of its pre-mean in the urban, more populous, higher-capacity, and richer counties.

The main finding is that a system-wide delivery-system privatization can deliver less measured healthcare per capita without a proportional loss in measured patient quality. The estimates do not vindicate the Hart et al. (1997) quality-deterioration corollary in this setting, since infant mortality fell where the reform had the most bite, nor the supply-stimulus prediction of the insurance-expansion literature (Dillender, 2022; Finkelstein, 2007;

Hackmann et al., 2025) in its unconditional form, since the local healthcare labor market contracted rather than expanded under the bundled treatment of capitation, single-network confinement, and ownership transfer. They rather support the claim that an ownership-and-payment change that constrains the per-enrollee budget can reduce per-capita inputs without proportional welfare losses when the pre-reform public system was underinvesting on the noncontractible quality margin (Hart et al., 1997; Propper, 2018). The budget record corroborates this, since the Commonwealth dissolved its public delivery agencies and routed financing through Medicaid managed care, whose outlay scaled to about $1.9 billion in 2025 dollars by fiscal year 2000 even as the per-capita inputs to the local healthcare sector fell (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1999, 2002). A back-of-the-envelope bound puts the short-horizon welfare gain at about $844 million in 2025 dollars over the four-year window on the avoided cost alone, the binding source of welfare improvement even when patient health is valued at zero (Section 6.4); an illustrative monetization of the subgroup infant-mortality improvement would raise the ceiling to about $4.0 billion but rests on the favorable subgroup and an offsetting maternal increase, so we do not rely on it. The same cost-benefit accounting that Duggan et al. (2023) use to find single-hospital privatization socially inefficient yields a positive sign here, because the inputs the system shed stood against a patient-health margin that improved rather than deteriorated.

For the privatization debates, the contribution is the counterfactual itself. The reforms that inspire those debates each move a partial bundle of the three margins Puerto Rico moved together: the Department of Veterans Affairs MISSION Act added outside-option access without transferring ownership, the English internal market introduced competition without capitating public delivery, and risk-based Medicaid managed care capitates payment without transferring facilities. Puerto Rico is the only case in which the bundle is complete and the delivery system itself changes ownership, so it speaks to the public-versus-private-delivery question the payer-side and single-facility literatures cannot reach. Those settings change who pays while providers stay private (Chorniy et al., 2018; Curto et al., 2019; Dranove et al., 2021; Duggan et al., 2018; Wallace, 2023), or compare public and private provision within a system where both exist (Chan et al., 2023), or convert a single facility within an intact market (Duggan et al., 2023); none observes the transition from public delivery system to private. Puerto Rico did so through procurement to a single plan per region, the market design Cuesta and Tebaldi (2025) show can raise welfare relative to regulated competition by removing cream-skimming.

The findings that this case establishes should apply to other settings. Contracting an entire publicly operated delivery network to private providers under capitation should be expected to contract per-capita inputs to the local healthcare sector. Therefore, moving a system like Medicaid managed care toward direct government delivery would expand that sector without an assured decrease in the death rate of patients (i.e. our measure for patient health). The comparison here runs in favor of the private model on resource use, because Puerto Rico's public system carried more inputs per resident without delivering better measured outcomes; whether welfare improves depends on the pre-reform system's resourcing, the local provider-mobility option, and the incidence of the contraction.

9 Tables

Table 1: County Descriptive Statistics by Year

	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001
Panel A: QCEW — Healthcare Labor Market †												
Employment per 1k pop.	2.86	3.28	3.44	3.61	3.90	4.03	4.54	5.02	5.52	5.92	6.22	7.67
	(4.74)	(5.15)	(5.05)	(5.23)	(5.41)	(5.33)	(5.56)	(5.85)	(6.49)	(6.76)	(7.16)	(9.94)
Establishments per 1k pop.	0.62	0.66	0.73	0.73	0.75	0.82	0.89	0.93	0.99	1.07	1.11	1.13
	(0.54)	(0.54)	(0.57)	(0.58)	(0.57)	(0.60)	(0.62)	(0.64)	(0.65)	(0.67)	(0.68)	(0.69)
Wage bill per 1k pop. ($)	5,405	6,558	7,059	7,454	7,966	8,478	9,490	10,651	11,852	12,772	13,439	16,981
	(10,809)	(12,859)	(13,280)	(13,985)	(13,962)	(14,211)	(15,151)	(16,101)	(18,296)	(19,145)	(20,646)	(26,925)
Counties	59	59	59	59	59	59	59	59	59	59	59	59
Treated Counties	–	–	–	–	5	16	38	38	58	58	59	59
Panel B: CMS — Provider of Services ‡												
Total Providers	–	202	222	242	248	254	259	262	265	268	274	275
Government	–	42	42	48	48	48	44	44	44	45	45	43
Private	–	160	180	194	200	206	215	218	221	223	229	232
Providers per 1k pop.	–	0.08	0.09	0.09	0.09	0.10	0.10	0.10	0.10	0.10	0.10	0.10
	(–)	(0.05)	(0.06)	(0.06)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)
Beds per 1k pop.	–	3.17	3.15	3.21	3.18	3.18	3.16	3.10	3.08	3.18	3.20	3.21
	(–)	(3.08)	(3.06)	(3.06)	(3.04)	(3.09)	(3.14)	(3.09)	(3.07)	(3.28)	(3.26)	(3.26)
Services per 1k pop.	–	0.46	0.48	0.48	0.49	0.49	0.49	0.48	0.47	0.48	0.48	0.49
	(–)	(0.47)	(0.49)	(0.49)	(0.49)	(0.50)	(0.49)	(0.47)	(0.47)	(0.47)	(0.47)	(0.51)
FTE per 1k pop.	–	9.04	9.17	9.54	12.85	12.54	12.41	12.33	12.17	12.10	12.36	12.61
	(–)	(8.34)	(8.44)	(8.45)	(19.54)	(19.46)	(18.40)	(18.09)	(17.96)	(18.14)	(18.96)	(19.14)
Counties	–	40	40	40	40	40	40	40	40	40	40	40
Treated Counties	–	–	–	–	5	10	29	29	39	39	40	40
Panel C: VS — Demographic Flow, Hospital Ownership, and Patient Health ¶												
Population	45,228	45,502	45,884	46,435	47,253	47,684	47,863	48,792	49,147	49,571	48,935	49,228
	(59,205)	(59,564)	(60,064)	(60,778)	(61,282)	(59,393)	(58,842)	(59,597)	(59,858)	(60,151)	(59,129)	(58,987)
Crude birth rate per 1k pop.	19.38	18.45	18.54	18.66	18.08	17.46	17.42	17.42	16.19	15.69	15.64	14.64
	(2.38)	(2.01)	(2.24)	(1.98)	(1.86)	(2.11)	(2.32)	(2.19)	(1.94)	(2.21)	(1.44)	(1.48)
Crude death rate per 1k pop.	7.02	6.97	7.19	7.43	7.28	7.67	7.57	7.44	7.55	7.23	7.16	7.21
	(1.14)	(1.01)	(1.01)	(1.06)	(1.20)	(1.31)	(1.28)	(1.19)	(1.15)	(1.10)	(1.11)	(1.00)
Crude net migration rate per 1k pop.	–	-5.51	-3.04	0.66	8.56	10.90	-2.87	13.48	-1.47	1.91	-9.89	1.04
	(–)	(2.28)	(2.72)	(2.51)	(29.54)	(67.48)	(26.57)	(20.11)	(17.10)	(17.97)	(78.74)	(8.35)
Government hospital birth share (%)	65.8	65.9	65.3	64.3	63.2	59.1	51.6	45.2	36.6	33.6	30.6	32.1
	(11.3)	(11.2)	(10.9)	(10.7)	(10.7)	(12.1)	(15.2)	(15.3)	(14.6)	(18.6)	(19.8)	(22.7)
Private hospital birth share (%)	33.8	33.8	34.4	35.5	36.5	39.6	48.1	54.6	63.1	66.2	69.2	67.8
	(11.4)	(11.3)	(11.0)	(10.7)	(10.9)	(11.6)	(15.3)	(15.3)	(14.6)	(18.6)	(19.9)	(22.8)
Infant mortality rate per 1k live births	12.74	12.67	12.71	13.40	11.84	14.28	9.91	12.10	10.28	10.79	10.15	9.62
	(6.09)	(5.30)	(4.47)	(5.45)	(4.97)	(7.33)	(4.90)	(6.42)	(5.13)	(4.30)	(5.45)	(5.04)
Low birth weight rate (%)	9.28	9.25	9.48	9.64	9.85	10.12	10.49	11.06	10.98	11.36	10.96	11.48
	(1.62)	(1.79)	(2.02)	(1.66)	(1.43)	(1.90)	(1.90)	(2.32)	(2.24)	(2.46)	(2.48)	(4.07)
Counties	78	78	78	78	78	78	78	78	78	78	78	78
Treated Counties	–	–	–	–	10	22	52	52	77	77	78	78
Panel D: PR DOL — Macro Labor Market #												
Labor force / pop. (%; Figure 3)	30.5	31.4	32.0	32.2	31.4	31.8	32.4	32.5	32.3	31.3	31.5	30.9
	(4.6)	(4.7)	(5.1)	(5.1)	(4.8)	(4.9)	(5.1)	(5.0)	(5.2)	(5.2)	(4.4)	(4.3)
Unemployment rate (%; U / LF)	16.9	18.4	19.3	20.1	17.1	16.0	15.8	16.1	15.8	14.2	11.6	13.1
	(4.7)	(4.9)	(5.1)	(5.1)	(4.6)	(4.7)	(4.5)	(4.8)	(4.6)	(4.7)	(2.2)	(2.7)
Counties	78	78	78	78	78	78	78	78	78	78	78	78
Treated Counties	–	–	–	–	10	22	52	52	77	77	78	78

Means with standard deviations in parentheses. Sample is the PR balanced-panel counties (78 total); per-dataset counts below). † *Panel A (QCEW):* Overall Healthcare = NAICS 621 (Ambulatory) + 622 (Hospitals) + 623 (Nursing/Residential). All wage variables are CPI-deflated using 1982-1984 as the base year. ‡ *Panel B (CMS):* Provider counts are point-in-time totals from the CMS Provider of Services file; Government / Private split is the contemporaneous ownership classification. The trajectory rows (Always Government, Always Private, Gov. → Private switchers) classify each provider over its full panel history and are stable across years. CMS balanced panel 1991–2000: 40 counties.*CMS provider mix (provider-years, 1991–2000):* The CMS county panel summarizes 2,586 provider-year observations from 296 unique facilities. By ownership: 454 (17.6%) government, 872 (33.7%) private for-profit, 1,024 (39.6%) private nonprofit, 236 (9.1%) private-unspecified. By facility category, 36.8% are Hospitals (CMS POS code 01); the remaining provider categories are Hospital (36.8%); Home Health Agency (19.8%); Hospice (18.1%); ESRD Facility (11.9%); Ambulatory Surgical Center (7.8%); Other (5.6%). ¶ *Panel C (VS):* Crude birth rate is live births per 1,000 mid-year population. Crude death rate is deaths per 1,000 mid-year population. Crude net migration rate is the residual of the demographic-balancing equation: $[(\text{Pop}_t - \text{Pop}_{t-1}) - (\text{Births}_t - \text{Deaths}_t)]/\text{Pop}_{t-1} \times 1000$, NA in 1990 (no $t-1$ population). Government and private hospital birth shares are percentages of all live births. Low birth weight rate is the percentage of live births under 2,500g. VS balanced panel 1990–2001: 78 counties. [2pt]# *Panel D (PR DOL):* Labor force / pop. is labor force divided by total county population × 100; employment / pop. is employment divided by total county population × 100. Both use the non-standard denominator (total population) rather than the BLS-standard civilian non-institutional population aged 16+ (CNIWAP), which is not available at the county level for the 1990–2001 panel. The unemployment rate is reported as BLS-standard (unemployed / labor force × 100). PR DOL balanced panel 1990–2001: 78 counties. Across all 936 county-year observations: BLS-style U/LF mean = 16.19%; alternative U/Pop mean = 5.04%; difference = 11.16 pp; Pearson correlation between the two series = 0.878. The two series differ in level but not in trends. *Pre-reform (1990–1993) cross-county medians used as thresholds in heterogeneity figures (Main Figure 5; Appendix Figures A.5–A.7; Online Appendix Figures OA.21–OA.22):* Government hospital birth share = 63.2%; Private hospital birth share = 36.3%; Healthcare employment per 1k pop. = 1.51; Population = 30,362; Unemployment rate = 18.6%. Each median is computed across counties after first averaging the outcome over the 1990–1993 window within each county.

Table 2: Pre-Reform Balance Across Treatment Cohorts (1990–1993 Means)

Variable	Cohort 1 (1994) *Mean*	Cohort 2 (1995) *Mean*	Cohort 3 (1996) *Mean*	Cohort 4 (1998) *Mean*	Cohort 5 (2000) *Mean*	F-test *p-value*	N
Total population	43,889	34,467	37,871	45,489	443,627	<0.001	78
Log population	10.124	10.287	10.330	10.427	13.003	0.013	78
Share urban (top-20 by 1990 density)	0.600	0.417	0.433	0.560	1.000	0.638	78
Gov. hospital birth share (%)	73.57	66.70	67.17	59.30	61.64	0.003	78
HC employment per 1k pop.	4.01	2.28	2.64	2.87	29.72	<0.001	60
Labor force participation rate (%)	34.96	29.79	28.96	33.90	35.47	<0.001	78
Unemployment rate (%)	15.23	19.60	20.97	17.12	10.70	<0.001	78
Crude net migration rate (per 1k)	-3.80	-2.87	-3.25	-1.48	1.89	0.002	78
Infant mortality rate (per 1k)	13.59	11.21	13.74	12.10	13.02	0.139	78
Crude birth rate (per 1k)	19.90	19.11	19.39	17.43	17.22	<0.001	78
Agriculture, forestry & fishing emp. per 1k	1.451	16.195	8.422	15.991	–	0.393	25
Transportation & warehousing emp. per 1k	3.595	0.623	0.644	1.287	9.811	0.037	78
Arts, entertainment & recreation emp. per 1k	4.745	–	0.281	0.342	2.145	0.525	11
Counties	10	12	30	25	1		78

Cell entries are 1990–1993 means of pre-reform covariates by treatment cohort. Cohorts 1–5 correspond to reform-rollout years 1994, 1995, 1996, 1998, and 2000 respectively (no rollouts in 1997 or 1999). The F-test column reports the p-value of the one-way ANOVA test for the joint null hypothesis that all five cohort means are equal (`aov(var ~ factor(cohort))`); a small p-value indicates that the listed covariate differs systematically across cohorts and is therefore not balanced under the rollout schedule. **Sources.** Population, gov. hospital birth share, infant mortality, crude birth rate, and crude net migration rate are 1990–1993 means from the Puerto Rico Department of Health vital statistics. Healthcare employment per 1,000 residents (NAICS Overall Healthcare = 621 + 622 + 623) and the three placebo-industry rows (Agriculture/Forestry/Fishing = NAICS 11; Transportation & Warehousing = NAICS 48-49; Arts/Entertainment/Recreation = NAICS 71) are 1990–1993 means computed from the QCEW (BLS) county-quarter panel: aggregated across ownership classes, averaged to annual frequency requiring at least 3 reporting quarters per year, then divided by the population. Labor force participation and unemployment rates are 1990–1993 means from the PR Department of Labor. The urban indicator equals one for the 20 counties with the highest 1990 population density and zero otherwise; the cohort-cell entry is the share of urban counties within the cohort. The three placebo-industry rows (NAICS 11, 48-49, 71) are the same industries used in the QCEW placebo robustness analysis (Online Appendix Section 7)

Table 3: Effects of the Reform on Hospital Ownership Composition (First Stage)

	(1) Gov. Hospital Birth Share (%)	(2) Priv. Hospital Birth Share (%)
ATT	-14.79***	15.19***
	(1.94)	(1.93)
Pre-treatment mean of dep. var.	62.55	37.04
ATT as % of pre-treatment mean	-23.6%	41.0%
N counties	78	78

Estimates use the Callaway and Sant'Anna (2021) ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$, based on two-sided tests with cluster-robust standard errors. The pre-treatment mean of the dependent variable is computed across treated counties using observations from the cohort-specific pre-treatment period (event time < 0). Government and private hospital birth shares are defined as the count of live births occurring at facilities of each ownership class divided by the total count of live births in the county-year, expressed in percent. The two shares are not exact complements because a small tail of births (typically <1%) is recorded as non-hospital or with unspecified facility ownership.

Table 4: Reform Effects on Healthcare Employment, Wages, and Establishments

	Healthcare (NAICS 621+622+623)			NAICS 621 (Ambulatory)		
	(1) Empl. per 1k	(2) Wages/Empl. per 1k	(3) Estabs. per 1k	(4) Empl. per 1k	(5) Wages/Empl. per 1k	(6) Estabs. per 1k
ATT	-0.511***	0.630	-0.050***	-0.502***	0.867	-0.034*
	(0.157)	(3.734)	(0.019)	(0.125)	(3.682)	(0.018)
Pre-treatment mean of dep. var.	3.923	38.962	0.759	2.895	38.889	0.747
ATT as % of pre-treatment mean	-13.0%	1.6%	-6.6%	-17.3%	2.2%	-4.6%
N counties	59	59	59	59	59	59

Estimates use the Callaway and Sant'Anna (2021) ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$, based on two-sided tests with cluster-robust standard errors. The pre-treatment mean of the dependent variable is computed across treated counties using observations from the cohort-specific pre-treatment period (event time < 0). The reference period for event-time aggregation is $t = -1$ (year before treatment).

Table 5: Effects of The Reform on Patient Health (Short-Horizon) and Hospital Capacity

	Patient Health (VS)		Hospital Capacity (CMS)	
	(1) IMR (per 1k live births)	(2) Low Birth Weight (%)	(3) Beds per 1k pop.	(4) Services per 1k pop.
ATT	-1.205	0.060	-0.265***	-0.024**
	(0.928)	(0.230)	(0.065)	(0.010)
Pre-treatment mean of dep. var.	12.620	9.717	3.389	0.487
ATT as % of pre-treatment mean	-9.5%	0.6%	-7.8%	-4.9%
N counties	78	78	40	40

Estimates use the Callaway and Sant'Anna (2021) ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$, based on two-sided tests with cluster-robust standard errors. The pre-treatment mean of the dependent variable is computed across treated counties using observations from the cohort-specific pre-treatment period (event time < 0). The reference period for event-time aggregation is $t = -1$ (year before treatment).

Table 6: Worker-Side Adjustment Margins to The Reform

	(1) Labor force / pop. (%)	(2) Crude Net Migration (per 1k)	(3) Crude Birth Rate (per 1k)
ATT	-1.110***	-14.651**	-0.564**
	(0.241)	(7.389)	(0.225)
Pre-treatment mean of dep. var.	31.895	1.059	18.267
ATT as % of pre-treatment mean	-3.5%	-1382.9%	-3.1%
N counties	78	78	78

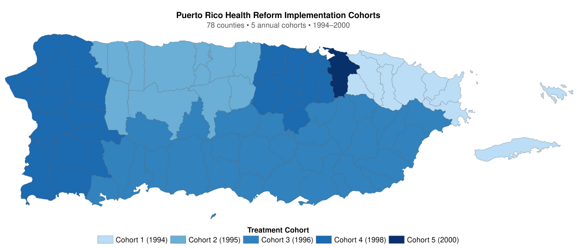
Estimates use the Callaway and Sant'Anna (2021) ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$, based on two-sided tests with cluster-robust standard errors. The pre-treatment mean of the dependent variable is computed across treated counties using observations from the cohort-specific pre-treatment period (event time < 0). The reference period for event-time aggregation is $t = -1$ (year before treatment).

Table 7: Heterogeneity in the Effects by Pre-Reform Public-Hospital Exposure

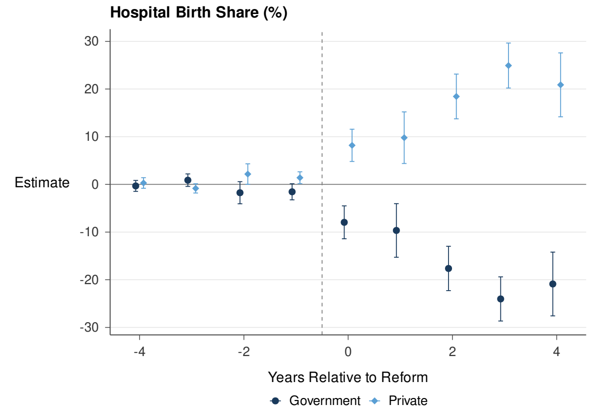
	(1) Gov. Hosp. Birth Share (%)	(2) HC Empl. per 1k	(3) Crude Net Migr. (per 1k)	(4) IMR (per 1k live births)
Below-median pre-reform government hospital birth share				
ATT	-15.49***	-0.346	-9.206	-1.454
	(2.44)	(0.233)	(10.814)	(1.101)
Pre-treatment mean	55.49	4.911	2.772	11.577
ATT as % of pre-mean	-27.9%	-7.0%	-332.1%	-12.6%
N counties	39	36	39	39
Above-median pre-reform government hospital birth share				
ATT	2.41	0.145	-23.891**	0.660
	(9.15)	(0.268)	(12.047)	(1.329)
Pre-treatment mean	71.21	2.069	-1.126	13.925
ATT as % of pre-mean	3.4%	7.0%	2121.2%	4.7%
N counties	39	23	39	39

Notes: Estimates use the Callaway and Sant'Anna (2021) doubly-robust ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$, based on two-sided tests with cluster-robust standard errors. counties are partitioned at the cross-county median of the 1990–1993 pre-reform government hospital birth share (median = 63.24%). *Below-median* (39 counties) = lower pre-reform reliance on the public hospital system; *Above-median* (39 counties) = higher pre-reform reliance.

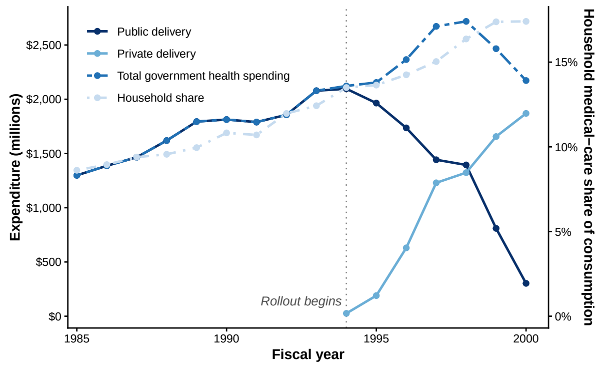
10 Figures



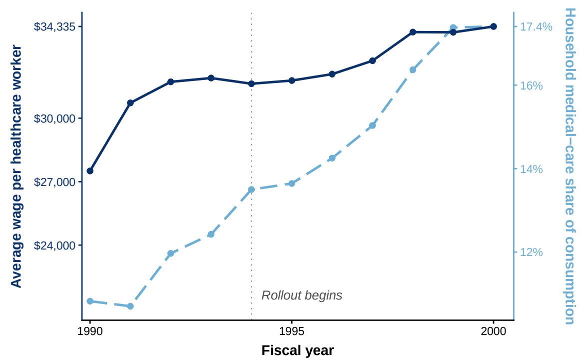
(a) Cohorts



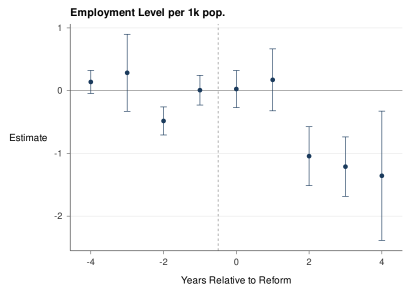
(b) Birth Share



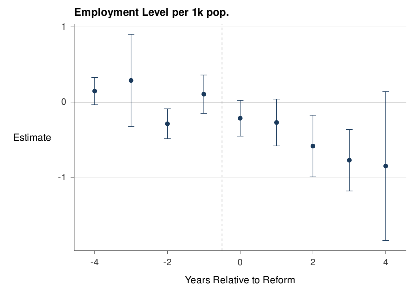
(c) Cost



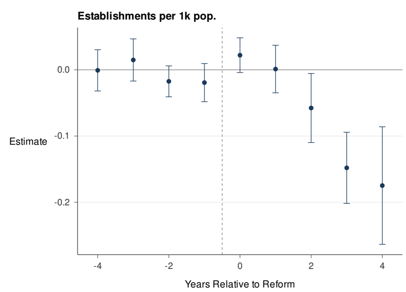
(d) Wages



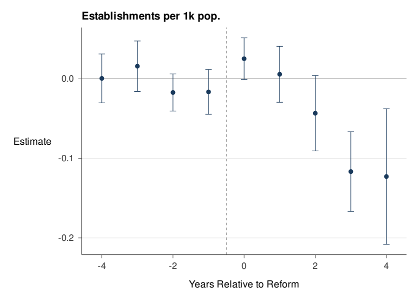
(a) Employment (per 1k) - 621+622+623



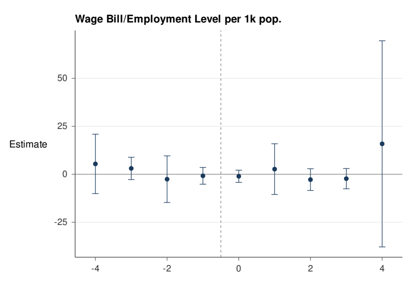
(b) Employment (per 1k) - 621



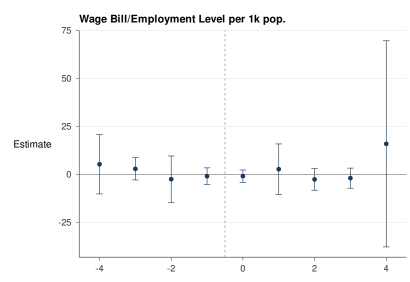
(c) Establishments (per 1k) - 621+622+623



(d) Establishments (per 1k) - 621

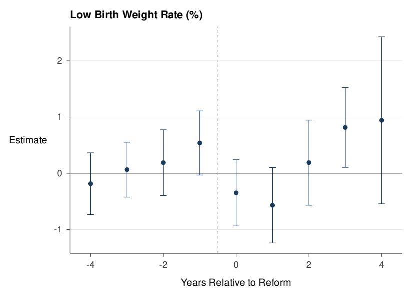


(c) Wage Bill/Employee (per 1k) - 621+622+623

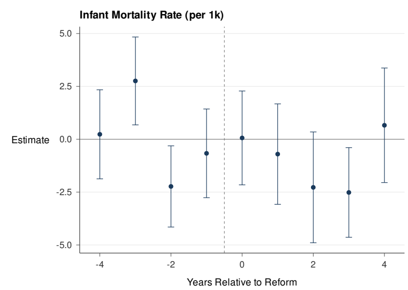


(d) Wage Bill per Employee (per 1k) - 621

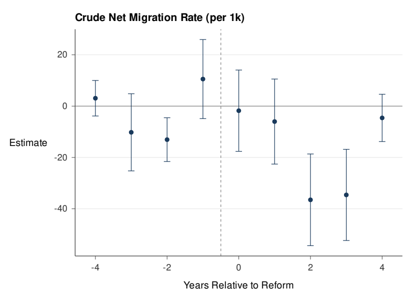
Figure 2: Healthcare Labor-Market Contraction



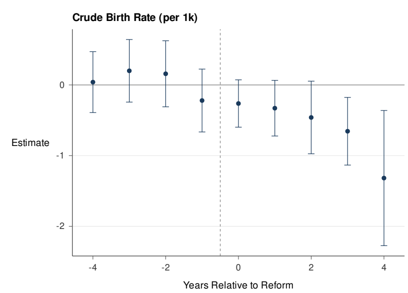
(a) Low Birth Weight



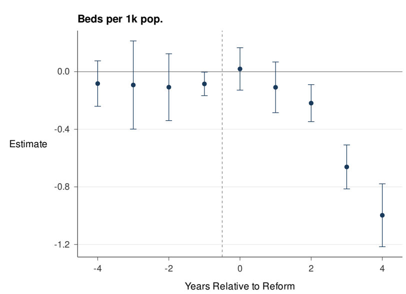
(b) Infant Mortality Rate



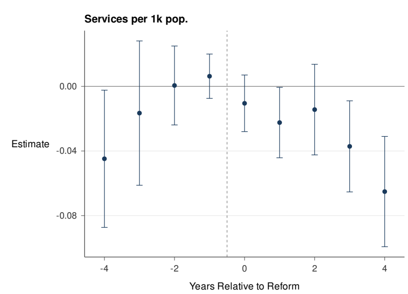
(c) Net Migration Rate



(d) Birth Rate

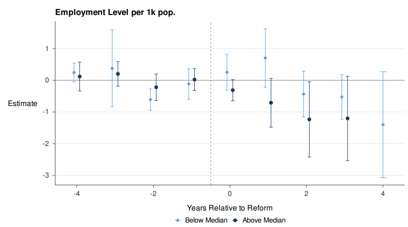


(e) Beds

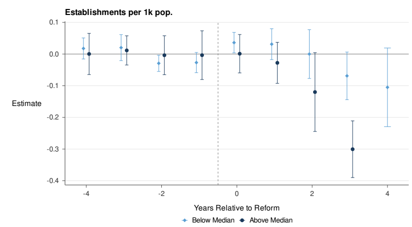


(f) Services

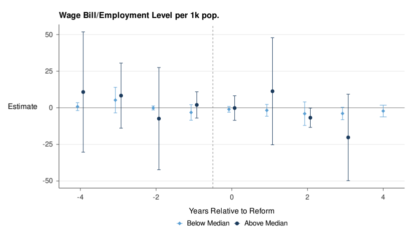
Figure 3: Patient Health



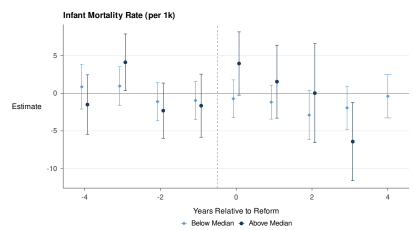
(a) Employment (per 1k) - 621+622+623



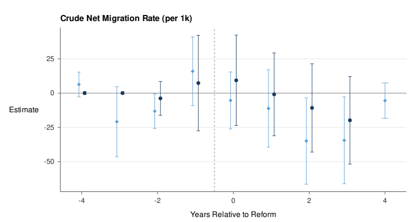
(b) Establishments (per 1k) - 621+622+623



(c) Wage Bill per Employee (per 1k) - 621+622+623

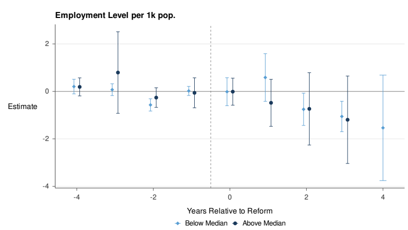


(d) Infant Mortality Rate (per 1k)

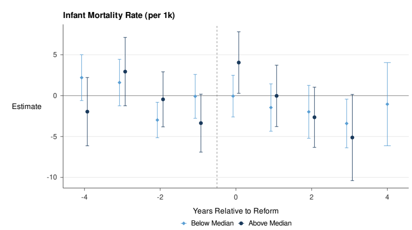


(d) Net Migration Rate

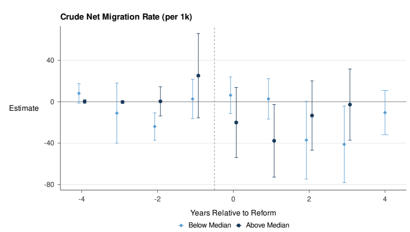
Figure 4: Heterogenous Effects by Poverty Rate



(a) Employment (per 1k) - 621+622+623



(b) Infant Mortality Rate (per 1k)



(c) Net Migration Rate (per 1k)

Figure 5: Heterogeneous Effects by Pre-Reform Gov. Hospital Exposure

References

Agafiev Macambira, D., Geruso, M., Lollo, A., Ndumele, C. D., & Wallace, J. (2026). The private provision of public services: Evidence from random assignment in Medicaid. American Economic Review, 116(6), 2038–2084. doi: 10.1257/aer.20230541

Aizer, A., Currie, J., & Moretti, E. (2007). Does managed care hurt health? evidence from Medicaid mothers. Review of Economics and Statistics, 89(3), 385–399. doi: 10.1162/rest.89.3.385

Arbona, G., & Ramirez de Arellano, A. (1978). Regionalization of health services: The Puerto Rican experience. New York: Oxford University Press. (Published for the International Epidemiological Association and the World Health Organization)

Arrow, K. J. (1963). Uncertainty and the welfare economics of medical care. American Economic Review, 53(5), 941–973.

Autor, D. H., Dorn, D., & Hanson, G. H. (2013). The China syndrome: Local labor market effects of import competition in the United States. American Economic Review, 103(6), 2121–2168. doi: 10.1257/aer.103.6.2121

Bloom, N., Propper, C., Seiler, S., & Van Reenen, J. (2015). The impact of competition on management quality: Evidence from public hospitals. Review of Economic Studies, 82(2), 457–489. doi: 10.1093/restud/rdu045

Borusyak, K., Jaravel, X., & Spiess, J. (2024). Revisiting event-study designs: Robust and efficient estimation. Review of Economic Studies, 91(6), 3253–3285. doi: 10.1093/restud/rdae007

Brugueras Fabre, A. J. (2002). Los costos de la reforma de salud de Puerto Rico. (Cited at p. 17 for the eight-region rollout date schedule)

Callaway, B., & Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. Journal of Econometrics, 225(2), 200–230. doi: 10.1016/j.jeconom.2020.12.001

Centers for Disease Control and Prevention. (2025). Behavioral risk factor surveillance system: Annual survey data. National Center for Chronic Disease Prevention and Health Promotion. Retrieved from `https://www.cdc.gov/brfss/annual_data/annual_data.htm` (Puerto Rico adult microdata on self-reported insurance coverage; 1996–2001 available, 1996–2000 used. Accessed November 2025)

Centers for Medicare and Medicaid Services. (1995). Medicaid managed care enrollment report (Tech. Rep.). Baltimore, MD: U.S. Department of Health and Human Services.

Centers for Medicare and Medicaid Services. (1996). Medicaid managed care enrollment report (Tech. Rep.). Baltimore, MD: U.S. Department of Health and Human Services.

Centers for Medicare and Medicaid Services. (1997). Medicaid managed care enrollment report (Tech. Rep.). Baltimore, MD: U.S. Department of Health and Human Services.

Centers for Medicare and Medicaid Services. (1998). Medicaid managed care enrollment report (Tech. Rep.). Baltimore, MD: U.S. Department of Health and Human Services.

Centers for Medicare and Medicaid Services. (1999). Medicaid managed care enrollment report (Tech. Rep.). Baltimore, MD: U.S. Department of Health and Human Services.

Centers for Medicare and Medicaid Services. (2000). National summary of state Medicaid managed care programs as of June 30, 2000 (Tech. Rep.). Baltimore, MD: U.S. Department of Health and Human Services. Retrieved from `https://web.archive.org/web/20060928140505/http://www3.cms.hhs.gov/`

MedicaidDataSourcesGenInfo/Downloads/descstprograms00.pdf
Centers for Medicare and Medicaid Services. (2001). National summary of state Medicaid managed care programs as of June 30, 2001 (Tech. Rep.). Baltimore, MD: U.S. Department of Health and Human Services. Retrieved from https://web.archive.org/web/20060928140217/http://www3.cms.hhs.gov/MedicaidDataSourcesGenInfo/Downloads/descstprograms01.pdf
Centers for Medicare and Medicaid Services. (2002). National summary of state Medicaid managed care programs as of June 30, 2002 (Tech. Rep.). Baltimore, MD: U.S. Department of Health and Human Services. (First year of mental-health/substance-abuse carve-out to MBHOs in Puerto Rico)
Centers for Medicare and Medicaid Services. (2003). National summary of state Medicaid managed care programs as of June 30, 2003 (Tech. Rep.). Baltimore, MD: U.S. Department of Health and Human Services.
Centers for Medicare and Medicaid Services. (2004). National summary of state Medicaid managed care programs as of June 30, 2004 (Tech. Rep.). Baltimore, MD: U.S. Department of Health and Human Services. (First year including Medicare dual eligibles in Puerto Rico's Reforma program)
Centers for Medicare and Medicaid Services. (2005). National summary of state Medicaid managed care programs as of June 30, 2005 (Tech. Rep.). Baltimore, MD: U.S. Department of Health and Human Services.
Centers for Medicare and Medicaid Services. (2023a). Healthcare cost report information system (HCRIS): Hospital cost reports (forms CMS-2552-10 and CMS-2552-96). Public-use files; distributed by the National Bureau of Economic Research. Retrieved from https://www.nber.org/research/data/hcris-hosp (Hospital 2552-10 and 2552-96 cost-report data; source for the total-cost-to-wage-bill ratio in the welfare calculation. Last updated February 2023; accessed November 2025)
Centers for Medicare and Medicaid Services. (2023b). Impact file for the hospital inpatient prospective payment system (IPPS). NBER-compiled annual files (J. Roth); distributed by the National Bureau of Economic Research. Retrieved from https://www.nber.org/research/data/centers-medicare-medicaid-services-cms-impact-file-hospital-ipps (Hospital beds, casemix, wage index, DSH percentage, cost-to-charge ratio, and Medicare discharges; used from 1994 onward. Last updated August 2023; accessed November 2025) doi: 10.60592/r1gg-ma38
Centers for Medicare and Medicaid Services. (2025). Provider of services files. Public-use files; distributed by the National Bureau of Economic Research. Retrieved from https://www.nber.org/research/data/provider-services-files (NBER Q4-of-year provider records; 1991–2001 retrieved, 1991–2000 used, aggregated to the county-year. Last updated September 2025; accessed November 2025)
Centers for Medicare and Medicaid Services. (2026). Medicaid & CHIP in Puerto Rico: State overview. Online state-overview page. Retrieved from https://www.medicaid.gov/state-overviews/puerto-rico.html
Chan, D. C., Card, D., & Taylor, L. (2023, November). Is there a VA advantage? Evidence from dually eligible veterans. American Economic Review, 113(11), 3003–3043. doi: 10.1257/aer.20211638
Chorniy, A., Currie, J., & Sonchak, L. (2018). Exploding asthma and ADHD caseloads:

The role of Medicaid managed care. Journal of Health Economics, 60, 1–15. doi: 10.1016/j.jhealeco.2018.04.002
Commonwealth of Puerto Rico. (1993). Ley núm. 72 de 7 de septiembre de 1993: Ley de la Administración de Seguros de Salud de Puerto Rico. Laws of Puerto Rico. (Puerto Rico Health Insurance Administration Act)
Congressional Budget Office. (2021, October). The veterans community care program: Background and early effects (CBO Publication No. 57583). Congressional Budget Office. Retrieved from https://www.cbo.gov/publication/57583 (VHA spending on community care grew from $7.9 billion in 2014 to $17.6 billion in 2021 (in 2021 dollars); community care rose from approximately 12 to 20 percent of VHA's medical-care budget over the same period)
Congressional Research Service. (2025a, September). Department of veterans affairs FY2025 appropriations (CRS Report No. R48608). Congressional Research Service. Retrieved from https://www.congress.gov/crs-product/R48608 (VHA medical-care discretionary appropriations of $112.58 billion for FY2025; reports VHA as "the nation's largest integrated health care delivery system" with more than 1,700 sites of care)
Congressional Research Service. (2025b). Medicaid: An overview (Tech. Rep. No. R42640). Washington, DC: Congressional Research Service. Retrieved from https://www.congress.gov/crs_external_products/R/HTML/R42640.html (Federal Medicaid funding is an open-ended entitlement to the states, with no upper limit on the federal match a state may receive; Medicaid programs in the territories (American Samoa, Guam, the Commonwealth of the Northern Mariana Islands, Puerto Rico, and the U.S. Virgin Islands) are instead subject to annual federal capped funding. Accessed June 2026)
Cooper, Z., Gibbons, S., Jones, S., & McGuire, A. (2011). Does hospital competition save lives? Evidence from the English NHS patient choice reforms. Economic Journal, 121(554), F228–F260. doi: 10.1111/j.1468-0297.2011.02449.x
Cuesta, J. I., & Tebaldi, P. (2025). Public procurement vs. regulated competition in selection markets (Working Paper No. 34141). National Bureau of Economic Research. Retrieved from https://www.nber.org/papers/w34141
Curto, V., Einav, L., Finkelstein, A., Levin, J., & Bhattacharya, J. (2019). Health care spending and utilization in public and private Medicare. American Economic Journal: Applied Economics, 11(2), 302–332. doi: 10.1257/app.20170295
Cutler, D. M., & Meara, E. (2000). The technology of birth: Is it worth it? In A. M. Garber (Ed.), Frontiers in health policy research, volume 3 (pp. 33–68). Cambridge, MA: MIT Press. (National Bureau of Economic Research; issued as NBER Working Paper No. 7390)
Cutler, D. M., Rosen, A. B., & Vijan, S. (2006). The value of medical spending in the United States, 1960–2000. New England Journal of Medicine, 355(9), 920–927. doi: 10.1056/NEJMsa054744
Cutler, D. M., & Scott Morton, F. (2013). Hospitals, market share, and consolidation. Journal of the American Medical Association, 310(18), 1964–1970. doi: 10.1001/jama.2013.281675
Dillender, M. (2022). How do Medicaid expansions affect the demand for healthcare workers? evidence from vacancy postings. Journal of Human Resources, 57(4), 1350–1393.

Retrieved from https://jhr.uwpress.org/content/57/4/1350.short

Dranove, D., Kessler, D., McClellan, M., & Satterthwaite, M. (2003). Is more information better? the effects of 'report cards' on health care providers. Journal of Political Economy, 111(3), 555–588. doi: 10.1086/374180

Dranove, D., Ody, C., & Starc, A. (2021). A dose of managed care: Controlling drug spending in Medicaid. American Economic Journal: Applied Economics, 13(1), 170–197. doi: 10.1257/app.20190165

Duggan, M. (2004). Does contracting out increase the efficiency of government programs? evidence from Medicaid HMOs. Journal of Public Economics, 88(12), 2549–2572. doi: 10.1016/j.jpubeco.2003.08.003

Duggan, M., Gruber, J., & Vabson, B. (2018). The consequences of health care privatization: Evidence from Medicare Advantage exits. American Economic Journal: Economic Policy, 10(1), 153–186. doi: 10.1257/pol.20160068

Duggan, M., Gupta, A., Jackson, E., & Templeton, Z. S. (2023). The impact of privatization: Evidence from the hospital sector (Working Paper No. 30824). National Bureau of Economic Research. Retrieved from https://www.nber.org/papers/w30824

Duggan, M., & Hayford, T. (2013). Has the shift to managed care reduced Medicaid expenditures? evidence from state and local-level mandates. Journal of Policy Analysis and Management, 32(3), 505–535. doi: 10.1002/pam.21693

Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto. (1997). Presupuesto recomendado para el año fiscal 1997–98. Oficina del Gobernador, Estado Libre Asociado de Puerto Rico. (Sector Salud: Departamento de Salud, Administración de Facilidades y Servicios de Salud (AFASS), Administración de Seguros de Salud (ASES), Administración de Servicios Médicos (ASEM), Administración de Servicios de Salud Mental y Contra la Adicción (ASSMCA), and the Corporación del Centro Cardiovascular. Budget columns report FY1995 and FY1996 actual (gasto), FY1996–97 vigente, and FY1997–98 recomendado. Referred to in project notes as "FY1996–97" (the vigente year); document title and issuing office reconstructed from internal section pages (cover not in the scanned set). Accessed June 2026)

Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto. (1998). Presupuesto recomendado para el año fiscal 1998–99: Departamento sombrilla de salud (presupuesto consolidado por agencia y origen de recursos). Oficina del Gobernador, Estado Libre Asociado de Puerto Rico. Retrieved from https://presupuestosanteriores.pr.gov/af99/tocs/presup99.htm (Health "sombrilla" (umbrella) consolidated budget by agency and by fund source. Budget columns report FY1996 and FY1997 actual (gasto), FY1998 vigente, and FY1999 aprobado. The sombrilla groups six delivery and regulatory agencies—Administración de Facilidades y Servicios de Salud (AFASS), Administración de Servicios de Salud Mental y Contra la Adicción (ASSMCA), Administración de Servicios Médicos (ASEM), Corporación del Centro Cardiovascular, Departamento de Salud, and Salud Correccional—and EXCLUDES the Administración de Seguros de Salud (ASES, the purchaser), which is organized separately. Also reports total sombrilla positions (26,905 in 1996 to 18,062 in 1999). Retrieved from the OGP Presupuestos Anteriores archive. Accessed June 2026)

Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto. (1999).

Presupuesto recomendado para el año fiscal 1999–2000: Administración de servicios médicos de puerto rico (asem). Oficina del Gobernador, Estado Libre Asociado de Puerto Rico. Retrieved from https://presupuestosanteriores.pr.gov/af2000/ingles/sombrill/salud/090.htm (English-version Web page ("Administration of Medical Services of Puerto Rico"). Legal basis: Law No. 66 of June 22, 1978. Budget columns report FY1997 and FY1998 actual (expended), FY1999 vigente (current), and FY2000 approved. Documents the transfer of the Pediatrics Hospital and University Hospital (supra-tertiary hospitals) to ASEM in FY1998–99 and onward to the Department of Health in FY1999–2000. Retrieved from the Oficina de Gerencia y Presupuesto budget archive; the rightmost (FY2000 approved) column is truncated in the retrieved page. The consolidated FY1997–2000 by-agency operating-budget table used in the budget panel (file Presupuesto_00.pdf) was retrieved from the OGP Presupuestos Anteriores index page (presupuesto.pr.gov/Pages/PRESUPUESTOSANTERIORES.aspx). The ASES "Beneficiarios de la Tarjeta de Salud enrollment-and-cost figure (FY1994–2000) used in the analysis was retrieved from presupuestosanteriores.pr.gov/af2000/dhtmlpag/presupugood.htm. Accessed June 2026)

Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto. (2001). Presupuesto recomendado para el año fiscal 2001–2002: Gastos operacionales por agencia (todos los orígenes de recursos). Oficina del Gobernador, Estado Libre Asociado de Puerto Rico. Retrieved from https://presupuesto.pr.gov/Pages/PRESUPUESTOSANTERIORES.aspx (Consolidated operating budget by agency. Budget columns report FY1999 and FY2000 actual (asignado/gastado), FY2001 vigente, and FY2002 recomendado. Health-sector agencies include the Departamento de Salud, Administración de Seguros de Salud (ASES), Administración de Facilidades y Servicios de Salud (AFASS, reported at $0 from FY2001), Administración de Servicios de Salud Mental y Contra la Adicción (ASSMCA), Corporación del Centro Cardiovascular, and Salud Correccional; the Administración de Servicios Médicos (ASEM) no longer appears. Retrieved from the OGP Presupuestos Anteriores index page (presupuesto.pr.gov/Pages/PRESUPUESTOSANTERIORES.aspx). Accessed June 2026)

Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto. (2002). Presupuesto recomendado para el año fiscal 2002–2003: Gastos operacionales por agencia (todos los orígenes de recursos). Oficina del Gobernador, Estado Libre Asociado de Puerto Rico. Retrieved from https://presupuesto.pr.gov/Pages/PRESUPUESTOSANTERIORES.aspx (Consolidated operating budget by agency. Budget columns report FY2000 and FY2001 actual, FY2002 vigente, and FY2003 recomendado. Health-sector agencies as in the FY2002 volume (AFASS at $0; ASEM absent). Retrieved from the OGP Presupuestos Anteriores index page (presupuesto.pr.gov/Pages/PRESUPUESTOSANTERIORES.aspx). Accessed June 2026)

Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto. (2004). Presupuesto recomendado para el año fiscal 2004–2005: Gastos operacionales por agencia (todos los orígenes de recursos). Oficina del Gobernador, Estado Libre Asociado de Puerto Rico. Retrieved from https://presupuesto.pr.gov/Pages/PRESUPUESTOSANTERIORES.aspx (Consolidated operating budget by agency. Budget columns report FY2002 and FY2003 actual, FY2004 vigente, and FY2005 recomendado. Health-sector agencies: Departamento de

Salud, ASES, ASSMCA, Corporación del Centro Cardiovascular, Salud Correccional. Retrieved from the OGP Presupuestos Anteriores index page (presupuesto.pr.gov/Pages/PRESUPUESTOSANTERIORES.aspx). Accessed June 2026)

Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto. (2005). Presupuesto recomendado para el año fiscal 2005–2006: Gastos operacionales por agencia (todos los orígenes de recursos). Oficina del Gobernador, Estado Libre Asociado de Puerto Rico. Retrieved from `https://presupuesto.pr.gov/Pages/PRESUPUESTOSANTERIORES.aspx` (Consolidated operating budget by agency. Budget columns report FY2003 and FY2004 actual, FY2005 vigente, and FY2006 recomendado. Health-sector agencies as in the FY2005 volume. Retrieved from the OGP Presupuestos Anteriores index page (presupuesto.pr.gov/Pages/PRESUPUESTOSANTERIORES.aspx). Accessed June 2026)

Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto. (2007). Presupuesto recomendado para el año fiscal 2007–2008: Consolidado por tipo de gasto y agencia. Oficina del Gobernador, Estado Libre Asociado de Puerto Rico. Retrieved from `https://presupuesto.pr.gov/Pages/PRESUPUESTOSANTERIORES.aspx` (Consolidated budget by TYPE OF EXPENSE and agency. Budget columns report FY2005 and FY2006 actual, FY2007 asignado, and FY2008 recomendado. Each agency is split across expense-type sections (Gastos de Funcionamiento and premium/assignment categories), so agency totals require summing sections; for ASES this separates agency operating cost ($375–462M) from managed-care premium pass-through to private plans ($984M–1,187M). Retrieved from the OGP Presupuestos Anteriores index page (presupuesto.pr.gov/Pages/PRESUPUESTOSANTERIORES.aspx). Accessed June 2026)

Estado Libre Asociado de Puerto Rico, Oficina de Presupuesto y Gerencia. (1989). Presupuesto para el año fiscal 1989–90, tomo i. Oficina del Gobernador, Estado Libre Asociado de Puerto Rico. (Departamento de Salud y Administración de Facilidades y Servicios de Salud (Section 29). Library call no. CAR 351.72 P977p 1989-90 v.1. Accessed June 2026)

Estado Libre Asociado de Puerto Rico, Oficina de Presupuesto y Gerencia. (1993). Presupuesto consolidado para el año fiscal 1994. Oficina del Gobernador, Estado Libre Asociado de Puerto Rico. (Departamento de Salud y Administración de Facilidades y Servicios de Salud (Section 2100), Administración de Servicios Médicos (ASEM, Section 2110), and Corporación del Centro Cardiovascular de Puerto Rico y del Caribe (Section 2120). Budget columns report fiscal years 1991–92 (histórico), 1992–93 (vigente), and 1993–94 (recomendado). Tomo and library call no. not determined from the scanned section pages. Accessed June 2026)

Farmer, C. M. (2023). The promise and challenges of VA community care: Veterans' issues in focus. RAND Health Quarterly, 10(3), 9. Retrieved from `https://pmc.ncbi.nlm.nih.gov/articles/PMC10273892/` (VA spending on community care grew from $7.9 billion in 2014 to $18.5 billion in 2021; community care accounted for 44 percent of VA's health care services across care settings as of 2022 testimony to the House Veterans' Affairs Committee)

Fernández, J., García-Pérez, M., & Orozco-Alemán, S. (2023). Unraveling the Hispanic health paradox. Journal of Economic Perspectives, 37(1), 145–168. doi: 10.1257/jep.37.1.145

Finkelstein, A. (2007). The aggregate effects of health insurance: Evidence from the introduction of Medicare. Quarterly Journal of Economics, 122(1), 1–37. doi: 10.1162/qjec.122.1.1

Finkelstein, A., Hendren, N., & Luttmer, E. F. P. (2019). The value of Medicaid: Interpreting results from the Oregon health insurance experiment. Journal of Political Economy, 127(6), 2836–2874. doi: 10.1086/702238

Finkelstein, A., Hendren, N., & Shepard, M. (2019). Subsidizing health insurance for low-income adults: Evidence from Massachusetts. American Economic Review, 109(4), 1530–1567. doi: 10.1257/aer.20171455

Gaynor, M., Ho, K., & Town, R. J. (2015). The industrial organization of health-care markets. Journal of Economic Literature, 53(2), 235–284. doi: 10.1257/jel.53.2.235

Gaynor, M., Moreno-Serra, R., & Propper, C. (2013). Death by market power: Reform, competition, and patient outcomes in the National Health Service. American Economic Journal: Economic Policy, 5(4), 134–166. doi: 10.1257/pol.5.4.134

Geruso, M., Layton, T. J., & Wallace, J. (2023). What difference does a health plan make? Evidence from random plan assignment in Medicaid. American Economic Journal: Applied Economics, 15(3), 341–379. doi: 10.1257/app.20210843

Gobierno de Puerto Rico. (2006). Presupuesto consolidado: Gastos operacionales por agencia [consolidated budget: Operational expenses by agency]. Tablas estadísticas, fiscal years 1985–2008, Oficina de Gerencia y Presupuesto. Retrieved from https://presupuestosanteriores.pr.gov/ (Appropriations to ASES, AFASS, ASEM, the Cardiovascular Center, and the Department of Health by fiscal year; compiled from the consolidated-budget editions for fiscal years 2000–2006. Accessed June 2026)

Goodair, B., & Reeves, A. (2022). Outsourcing health-care services to the private sector and treatable mortality rates in England, 2013–20: An observational study of NHS privatisation. The Lancet Public Health, 7(7), e638–e646. doi: 10.1016/S2468-2667(22)00133-5

Grossman, S. J., & Hart, O. D. (1986). The costs and benefits of ownership: A theory of vertical and lateral integration. Journal of Political Economy, 94(4), 691–719. doi: 10.1086/261404

Gruber, J. (2017). Delivering public health insurance through private plan choice in the United States. Journal of Economic Perspectives, 31(4), 3–22. doi: 10.1257/jep.31.4.3

Hackmann, M. B., Heining, J., Klimke, R., Polyakova, M., & Seibert, H. (2025, January). Health insurance as economic stimulus? evidence from long-term care jobs (Working Paper No. 33429). National Bureau of Economic Research. Retrieved from https://www.nber.org/papers/w33429

Hart, O., Shleifer, A., & Vishny, R. W. (1997). The proper scope of government: Theory and an application to prisons. Quarterly Journal of Economics, 112(4), 1127–1161. doi: 10.1162/003355300555448

Instituto de Administración y Política de Salud. (1995). Coordinating health care reform with the U.S. territories and possessions: The case of Puerto Rico (Final Report No. Grant Number 18-C-90240/2-01). San Juan, Puerto Rico: University of Puerto Rico, Medical Sciences Campus. (Prepared for the Health Care Financing Administration)

Junta de Planificación de Puerto Rico. (2024). Informe económico al gobernador: Apéndice estadístico. Estado Libre Asociado de Puerto Rico, Oficina del Gobernador. Retrieved from https://jp.pr.gov/informe-economico-al-gobernador/ (Table 5, Personal Consumption Expenditures by Major Type of Product, Fiscal Years. The medical-care expenditure series used here is spliced across the 1989, 1999, 2005, 2008, 2015, 2022, and 2024 editions. Accessed June 2026)

Kaiser Family Foundation. (2024a). Medicare advantage in 2024: Enrollment update and key trends. KFF issue brief. Retrieved from https://www.kff.org/medicare/medicare-advantage-in-2024-enrollment-update-and-key-trends/ (In 2024, 32.8 million people (54% of eligible Medicare beneficiaries) were enrolled in Medicare Advantage, accounting for $462 billion (54%) of federal Medicare spending, up from $156 billion in 2014; CBO projects the Medicare Advantage share will reach 64% by 2034. Plans are paid a capitated amount per enrollee)

Kaiser Family Foundation. (2024b). Recent changes in Medicaid financing in Puerto Rico and other U.S. territories. KFF issue brief. Retrieved from https://www.kff.org/elections/recent-changes-in-medicaid-financing-in-puerto-rico-and-other-u-s-territories/

Kaiser Family Foundation. (2025). 10 things to know about Medicaid Managed Care. KFF issue brief, last updated October 2025. Retrieved from https://www.kff.org/medicaid/10-things-to-know-about-medicaid-managed-care/ (Reports FFY 2024 figures: 291 contracting MCOs across 42 states and DC; 66 million enrollees in comprehensive MCOs; capitated MCO payments equal to 50 percent of total Medicaid spending of $919 billion)

Kaiser Family Foundation. (2026). Medicaid financing: The basics. KFF issue brief, last updated March 2026. Retrieved from https://www.kff.org/medicaid/medicaid-financing-the-basics/ (FFY 2024 totals: $919 billion total Medicaid spending, comprising $594 billion federal (65%) and $325 billion state (35%); capitated MCO payments accounted for 50% of Medicaid spending)

King's Fund. (2026). The NHS budget and how it has changed. King's Fund data brief, last updated March 2026. Retrieved from https://www.kingsfund.org.uk/insight-and-analysis/data-and-charts/nhs-budget-nutshell (DHSC total spending of £204.7 billion in 2024/25, of which £187 billion was day-to-day spending allocated to NHS England for health services)

Marton, J., Yelowitz, A., & Talbert, J. C. (2014). A tale of two cities? the heterogeneous impact of Medicaid managed care. Journal of Health Economics, 36, 47–68. doi: 10.1016/j.jhealeco.2014.03.001

Medicaid and CHIP Payment and Access Commission. (2020, August). Medicaid and CHIP in Puerto Rico (Tech. Rep.). MACPAC. Retrieved from https://www.macpac.gov/wp-content/uploads/2020/08/Medicaid-and-CHIP-in-Puerto-Rico.pdf

Medicaid and CHIP Payment and Access Commission. (2026, February). MACStats: Medicaid and CHIP data book (Tech. Rep.). MACPAC. Retrieved from https://www.macpac.gov/publication/macstats-medicaid-and-chip-data-book/ (Total Medicaid spending of $957.4 billion in FY2024; 78.0 million Medicaid and CHIP enrollees as of July 2025 (over 31% of the U.S. population); federal share 64.7% of benefit spending; more than half of benefit spending is capitation to managed-care plans; Medicaid is 9.1% and Medicare 12.8% of federal outlays. Post-pandemic eligibility redeterminations reduced enrollment over 2023–2025)

Medicare Payment Advisory Commission. (2024, March). Report to the Congress: Medicare payment policy (Tech. Rep.). MedPAC. Retrieved from https://www.medpac.gov/wp-content/uploads/2024/03/Mar24_Ch12_MedPAC_Report_To_Congress_SEC.pdf (Estimates payments to Medicare Advantage plans average 122% of spending for comparable beneficiaries in traditional fee-for-service Medicare, an estimated $83 billion in higher spending in 2024 (Chapter 12, The Medicare Advantage program))

Megginson, W. L., & Netter, J. M. (2001). From state to market: A survey of empirical studies on privatization. Journal of Economic Literature, 39(2), 321–389. doi: 10.1257/jel.39.2.321

National Audit Office. (2026). Department of health and social care 2024-25: Overview. NAO Departmental Overview, January 2026. Retrieved from https://www.nao.org.uk/overviews/department-of-health-and-social-care-2024-25/ (DHSC spent £219.2 billion gross (£204.7 billion after income) in 2024-25; £24.089 billion was spent on the purchase of healthcare from non-NHS bodies)

National Center for Health Statistics. (2025). Vital statistics natality birth data. National Vital Statistics System; distributed by the National Bureau of Economic Research. Retrieved from https://www.nber.org/research/data/vital-statistics-natality-birth-data (U.S. Territory (PS) files; Puerto Rico birth records begin 1994. Last updated September 2025; accessed November 2025) doi: 10.60592/107a-5j74

Pan American Health Organization. (2007, September). Health systems profile: Puerto Rico—monitoring and analyzing health systems change/reform (Tech. Rep.). Washington, D.C.: Pan American Health Organization/World Health Organization. Retrieved from https://estadisticas.pr/files/BibliotecaVirtual/estadisticas/biblioteca/DS/DS_Perfil_Sistema_Salud-Puerto_Rico_2007_espanol_e_ingles.pdf

Propper, C. (2018). Competition in health care: Lessons from the English experience. Health Economics, Policy and Law, 13(3-4), 492–508. doi: 10.1017/S1744133117000494

Puerto Rico Department of Health. (2025). Annual vital statistics report (Informe Anual de Estadísticas Vitales), 1990–2001. Instituto de Estadísticas de Puerto Rico (Registro Demográfico). Retrieved from https://www.estadisticas.pr.gov/en-us/productos/informe-anual-estadisticas-vitales (Annual county vital records and resident population, 1990–2001; the canonical per-capita denominator. Accessed November 2025)

Puerto Rico Department of Labor and Human Resources. (2025). Estadísticas de desempleo por municipios (laus): Local area unemployment statistics, serie histórica. Instituto de Estadísticas de Puerto Rico. Retrieved from https://estadisticas.pr/files/Inventario/publicaciones/Serie%20Hist%C3%B3rica%20Natural_0.pdf (County (municipio) labor-force and unemployment series; full series 1990–2023, years 1990–2000 used. Accessed November 2025)

Shepard, M., & Wallace, J. (2026). Understanding Medicaid managed care: The procured competition model. Journal of Economic Perspectives, 40(2), 171–194. doi: 10.1257/jep.20251475

Shleifer, A. (1998). State versus private ownership. Journal of Economic Perspectives, 12(4), 133–150. doi: 10.1257/jep.12.4.133

U.S. Bureau of Labor Statistics. (2025). Quarterly census of employment and wages. Public-use CSV data files by industry, U.S. Department of Labor. Retrieved from https://www.bls.gov/cew/downloadable-data-files.htm (County employment, wages, and establishment counts by NAICS; full series 1990–2023, years 1990–2000 used. Accessed November 2025)

U.S. Census Bureau. (2000a). Census 2000 minor civil division worker flow files: Puerto Rico. Special tabulation, file 2kwrkmcd_pr.xls, U.S. Department of Commerce. Retrieved from https://www2.census.gov/programs-surveys/decennial/tables/2000/mcd-worker-flow-files/2kwrkmcd_pr.xls (Residence to workplace minor civil division/county worker flows; municipio-to-municipio commuting. Accessed June 2026)

U.S. Census Bureau. (2000b). Census 2000, summary file 4 (SF4), table PCT148. Decennial Census of Population and Housing, U.S. Department of Commerce; retrieved via data.census.gov. Retrieved from https://data.census.gov/table/DECENNIALSF42000.PCT148 ("Poverty Status in 1999 by Disability Status by Age" (Puerto Rico, county level); county poverty rate used as a heterogeneity covariate. Accessed November 2025)

U.S. Environmental Protection Agency. (2010). Guidelines for preparing economic analyses. National Center for Environmental Economics, Office of Policy. (Updated May 2014; establishes the recommended central value of a statistical life)

Viscusi, W. K., & Aldy, J. E. (2003). The value of a statistical life: A critical review of market estimates throughout the world. Journal of Risk and Uncertainty, 27(1), 5–76. doi: 10.1023/A:1025598106257

Wallace, J. (2023). What does a provider network do? evidence from random assignment in Medicaid managed care. American Economic Journal: Economic Policy, 15(1), 473–509. doi: 10.1257/pol.20210162

Wooldridge, J. M. (2025). Two-way fixed effects, the two-way Mundlak regression, and difference-in-differences estimators. Empirical Economics, 69, 2545–2587. doi: 10.1007/s00181-025-02807-z

World Health Organization, & Pan American Health Organization. (1998). La reforma del sector salud: El caso de Puerto Rico (Tech. Rep.). Washington, D.C.: Pan American Health Organization.

Yagan, D. (2019). Employment hysteresis from the Great Recession. Journal of Political Economy, 127(5), 2505–2558. doi: 10.1086/701809